



Operator's Manual SRM-225 Grass Trimmer / Brushcutter

WARNING

The engine exhaust from this product contains chemicals known to the State of California to cause cancer, birth defects, or other reproductive harm.

WARNING



Read and understand all provided literature before use. Failure to do so could result in serious injury.

Note: This product complies with CAN ICES-2/NMB-2.

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INTRODUCTION

Specifications, descriptions, and illustrative material in this literature are as accurate as possible. Specifications are subject to change without notice. Illustrations might include optional equipment and accessories, and might not include all standard equipment. Your equipment might appear slightly different than pictured equipment.



Read and understand all provided literature. Literature contains specifications and information for safety, operation, maintenance, storage, and assembly specific to this product. Scan QR codes for more information.



For additional literature, including safety manuals where applicable, or questions regarding terms used in this manual, visit:

<https://www.echo-usa.com/manuals>

OR

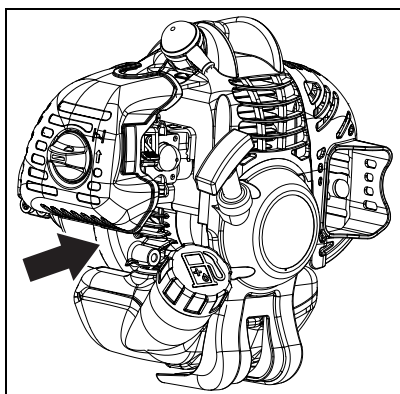


<https://www.shindaiwa-usa.com/manuals>

SERVICING INFORMATION

Parts and Serial Number

Genuine ECHO parts and assemblies for your ECHO products are available only from an Authorized ECHO Dealer. When you do need to buy parts always have the model number and serial number of the unit with you. For future reference write them in the space provided below.



Model No. _____ Serial No. _____

Service

Service of this product during the warranty period must be performed by an Authorized ECHO Service Dealer. For the name and address of the Authorized ECHO Service Dealer nearest you, ask your retailer or call: 1-800-432-ECHO (3246). Dealer information is also available on our Web Site www.echo-usa.com. When presenting your unit for Warranty service/repairs, proof of purchase is required.

ECHO Consumer Product Support

If you require assistance or have questions concerning the application, operation, or maintenance of this product, call the ECHO Consumer Product Support Department at 1-800-432-ECHO (3246) from 8:00 a.m. to 5:00 p.m. (Central Standard Time) Monday through Friday. Before calling, please know the model and serial number of your unit.

Product Registration

Register your ECHO equipment online at www.echo-usa.com or by filling out the product registration sheet included in this manual. Registering your product confirms warranty coverage and provides a direct link to ECHO if we find it necessary to contact you.

Additional Literature

In addition to finding information online, information is available from your Authorized ECHO Service Dealer, or by contacting ECHO Incorporated, 400 Oakwood Road, Lake Zurich, IL 60047, 1-800-432-ECHO (3246).

SAFETY

Manual Safety Symbols and Important Information

Throughout this manual and on the product itself, you will find safety alerts and helpful, informational messages preceded by symbols or key words. The following is an explanation of those symbols and key words and what they mean to you.

DANGER

The safety alert symbol accompanied by the word “DANGER” calls attention to an act or condition which WILL lead to serious personal injury or death if not avoided.

WARNING

The safety alert symbol accompanied by the word “WARNING” calls attention to an act or condition which CAN lead to serious personal injury or death if not avoided.

CAUTION

The safety alert symbol accompanied by the word “CAUTION” calls attention to an act or condition which might lead to minor or moderate personal injury if not avoided.

NOTICE

The enclosed message provides information necessary for the protection of the unit.

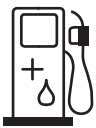
Note: This enclosed message provides tips for use, care and maintenance of the unit.



CIRCLE AND SLASH SYMBOL

This symbol means the specific action shown is prohibited. Ignoring these prohibitions can result in serious or fatal injury.

International Symbols

Symbol	Description	Symbol	Description
	Warning, See Operator's Manual	H	Carburetor Adjustment - High Speed Mixture
	Wear Eye, Ear and Head Protection	T	Carburetor Adjustment - Idle Speed
	Wear Hand and Foot Protection	L	Carburetor Adjustment - Low Speed Mixture
	Safety/Alert		STOP Switch
	Hot Surface		Fuel and Oil Mixture
	DO NOT Allow Flames or Sparks Near Fuel		Ignition ON / OFF
	DO NOT Smoke Near Fuel		Purge Bulb
	Choke Control "RUN" Position (Choke Open)		Choke Control "COLD START" Position (Choke Closed)
	Keep Feet Away From Blade		Rotating Cutting Attachment

Symbol	Description	Symbol	Description
	Thrown Objects		Direction of Blade
	DO NOT USE LINE HEADS - Blades Only		DO NOT USE BLADES - Line Heads Only
		AVOID KICKOUT Keep Bystanders at Least 15 m (50 ft.) Away	
		Beware Thrown Objects Wear Eye Protection	
		Keep Bystanders and Helpers Away 15 m (50 ft.)	

Note: Not all symbols will appear on your unit.



Personal Condition and Safety Equipment

WARNING

Cancer and Reproductive Harm
www.P65Warnings.ca.gov

WARNING



The muffler or catalytic muffler and surrounding cover may become extremely hot. Always keep clear of exhaust and muffler area, otherwise serious personal injury might occur.

WARNING

Users of this product risk injury to themselves and others if the unit is used improperly and/or safety precautions are not followed. Proper clothing and safety gear must be worn when operating unit.

Physical Condition

Your judgment and physical dexterity may not be good:

- If you are tired or sick
- If you are taking medication
- If you have taken alcohol or drugs

Operate unit only if you are physically and mentally well.

Eye Protection

WARNING

- ◆ Eye protection that meets ANSI Z87.1 or CE requirements must be worn whenever you operate the unit.
- ◆ For additional safety, a full-face shield can be worn over safety glasses or goggles to provide protection from sharp branches or flying debris.

Hand Protection

Wear sturdy, no-slip, rubber work gloves to improve your grip on the handles. Gloves also provide protection against cuts and scratches, cold environments, and reduce the transmission of machine vibration to your hands.

Hearing and Ear Protection

ECHO recommends wearing personal protective equipment whenever unit is used.

Breathing Protection

Operators who are sensitive to dust or other common airborne allergens may need to wear a dust mask to prevent inhaling these materials while operating unit. Dust masks can provide protection against dust, plant debris, and other plant matter such as pollen. Make sure the mask does not impair your vision, and replace the mask as needed to prevent air restrictions.

Proper Clothing

Wear snug-fitting, durable clothing:

- Pants should have long legs, shirts should have long sleeves.
- DO NOT WEAR SHORTS.
- DO NOT WEAR TIES, SCARVES, JEWELRY, or clothing with loose or hanging items that could become entangled in moving parts or surrounding growth.
- Keep clothing buttoned or zipped, and keep shirt tails tucked in.

Wear sturdy work shoes with nonskid rubber soles.

- DO NOT WEAR OPEN TOED SHOES.
- DO NOT OPERATE UNIT WITH BARE FEET.

Keep long hair away from engine and air intake. Retain hair with cap or net.

Heavy protective clothing can increase operator fatigue, which may lead to heat stroke. Schedule heavy work for early morning or late afternoon hours when temperatures are cooler.

 WARNING

The components of this machine generate an electromagnetic field during operation, which can interfere with some pacemakers. To reduce the risk of serious or fatal injury, persons with pacemakers should consult with their physician and the pacemaker manufacturer before operating this machine. In the absence of such information, ECHO does not recommend the use of this machine by anyone who has a pacemaker.

Extended Operation and Extreme Conditions **CAUTION**

Prolonged exposure to cold and/or vibration can result in injury. Read and follow all safety and operation instructions to minimize risk of injury. Failure to follow instructions can result in painful wrist/hand/arm injuries.

It is believed that a condition called Raynaud's Phenomenon, which affects the fingers of certain individuals, may be brought about by exposure to vibration and cold. Exposure to vibration and cold may cause tingling and burning sensations, followed by loss of color and numbness in the fingers. The following precautions are strongly recommended, because the minimum exposure, which might trigger the ailment, is unknown.

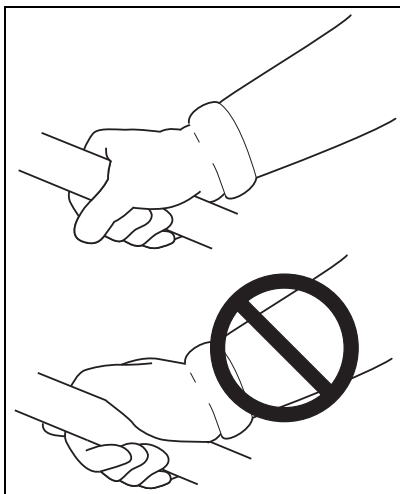
- Keep your body warm, especially the head, neck, feet, ankles, hands, and wrists.
- Maintain good blood circulation by performing vigorous arm exercises during frequent work breaks, and also by not smoking.
- Limit the hours of operation. Try to fill each day with jobs where operating the unit or other hand-held power equipment is not required.
- If you experience discomfort, redness, and swelling of the fingers followed by whitening and loss of feeling, consult your physician before further exposing yourself to cold and vibration.

Repetitive Stress Injuries (RSI)

It is believed that overusing the muscles and tendons of the fingers, hands, arms, and shoulders may cause soreness, swelling, numbness, weakness, and extreme pain in those areas. Certain repetitive hand activities may put you at a high risk for developing a Repetitive Stress Injury (RSI). An extreme RSI condition is Carpal Tunnel Syndrome (CTS), which could occur when your wrist swells and squeezes a vital nerve that runs through the area. Some believe that prolonged exposure to vibration may contribute to CTS. CTS can cause severe pain for months or even years.

To reduce the risk of RSI/CTS, do the following

- Avoid using your wrist in a bent, extended, or twisted position. Instead try to maintain a straight wrist position. Also, when grasping, use your whole hand, not just the thumb and index finger.
- Take periodic breaks to minimize repetition and rest your hands.
- Reduce the speed and force with which you do the repetitive movement.
- Do exercises to strengthen the hand and arm muscles.
- Immediately stop using all power equipment and consult a doctor if you feel tingling, numbness, or pain in the fingers, hands, wrists, or arms. The sooner RSI/CTS is diagnosed, the more likely permanent nerve and muscle damage can be prevented.



DANGER

All over head electrical conductors and communications wires can have electricity flow with high voltages. This unit is not insulated against electrical current. Never touch wires directly or indirectly, otherwise serious injury or death can result.

DANGER

Do not operate this product indoors or in inadequately ventilated areas. Engine exhaust contains poisonous emissions and can cause serious injury or death.

Read Manuals

- Provide all users of this equipment with literature for instructions on safe operation.

Clear the Work Area

- Always clear the work area of foreign objects such as rocks, broken glass, nails, wire, or string, and check for any hidden hazards. Spectators and fellow workers must be warned, and children and animals prevented from coming nearer than 15 m (50 ft.) while the unit is in use.
- Outside the 15 m (50 ft.) zone, there is still a risk of injury from thrown objects.
- Bystanders should be encouraged to wear eye protection.
- If you are approached, stop the engine and cutting attachment.
- When a bladed unit is used, there is the added risk of injury to bystanders being struck with the moving blade in the event of a blade thrust or other unexpected reaction of the blade.

Keep a Firm Grip

- Always hold throttle handle and support handle with thumbs and fingers tightly encircling the handles.

Keep a Solid Stance

- Maintain footing and balance at all times. Do not stand on slippery, uneven or unstable surfaces. Do not work in odd positions or on ladders. Do not over reach.
- Keep cutting attachment below waist.
- Keep all body parts away from rotating cutting attachment.

Avoid Hot Surfaces

- Keep exhaust area clear of flammable debris. Avoid contact during and immediately after operation.



Equipment

WARNING

Use only approved attachments. Serious injury may result from the use of a non-approved attachment combination. ECHO Incorporated will not be responsible for the failure of cutting devices, attachments or accessories which have not been tested and approved by ECHO Incorporated. Read and comply with all safety instructions.

- ◆ Do not attempt to modify this product. Serious injury can result from the use of any modified product.
- ◆ Check unit for loose or missing nuts, bolts, and screws. Tighten or replace as needed.
- ◆ Inspect shield for damage and ensure that shield is properly installed, and that the cut-off knife is securely in place. Replace if either is damaged or missing.
- ◆ Check that the cutting attachment is firmly attached and in safe operating condition.
- ◆ Manufacturer recommended flexible non-metallic line is installed in the trimmer head.
- ◆ Assure that throttle trigger, throttle trigger lockout, and stop switch all work properly.
- ◆ Check that handle and harness (if included) are installed and adjusted for safe, comfortable operation. See Assembly Section for proper adjustment.

WARNING

Moving parts can amputate fingers or cause severe injuries. Keep hands, clothing and loose objects away from all openings.

- ◆ ALWAYS stop engine, disconnect spark plug, and make sure all moving parts have come to a complete stop before assembling unit, removing obstructions, clearing debris, or servicing unit.
- ◆ Do not connect spark plug lead to spark plug until unit is ready for use.
- ◆ DO NOT start or operate unit unless all guards and protective covers are properly assembled to unit.
- ◆ NEVER reach into any opening while the engine is running. Moving parts may not be visible through openings.

- ◆ Position wiring safely to prevent snagging, separation of connectors, or breakage during operation. Gather excess wire, and secure with wiring clamp if provided on equipment, or tuck behind the air filter area. Do not place wiring directly against hot engine components.
- ◆ Check wiring and connectors for nicks, cuts, exposed wire, or other damage, and repair or replace as needed. Exposed wire or connectors can cause shocks, sparks, and risk of fire or explosion, resulting in serious injury.
- ◆ Check wire terminals for secure connections.

**WARNING**

Periodically Check fuel system (fuel lines, vent, grommet, fuel tank, and fuel cap) for leaks especially if the unit is dropped. If damage or leaks are found, do not use unit, otherwise serious personal injury or property damage may occur. Have unit repaired by an authorized servicing dealer before using.

EMISSION CONTROL (EXHAUST & EVAPORATIVE)

CARB and EPA Emissions Control Information

The emission control system for the engine is EM (engine modification) and, if the second to last character of the Engine Family on the Emission Control Information label (sample below) is "B", "C", "K", or "T", the emission control system is EM and TWC (3-way catalyst). The fuel tank/fuel line emission control system is EVAP (evaporative emissions). Evaporative emissions for California models are only applicable to fuel tanks and fuel feed lines.

An Emission Control Label is located on the engine. (This is an EXAMPLE ONLY, information on label varies by engine FAMILY).

EMISSION CONTROL INFORMATION
ENGINE FAMILY: EJERHS.0214EQ DISPLACEMENT: 21.2cc
EMISSION COMPLIANCE PERIOD: 50Hours
THIS ENGINE MEETS 2013 U.S. EPA EXH/EVP & CALIFORNIA
EXH/EVP EMISSION REGULATIONS FOR S.O.R.E. REFER TO OWNER'S
MANUAL FOR MAINTENANCE SPECIFICATIONS AND ADJUSTMENTS.
YAMABIKO CORP. MMM/YYYY 

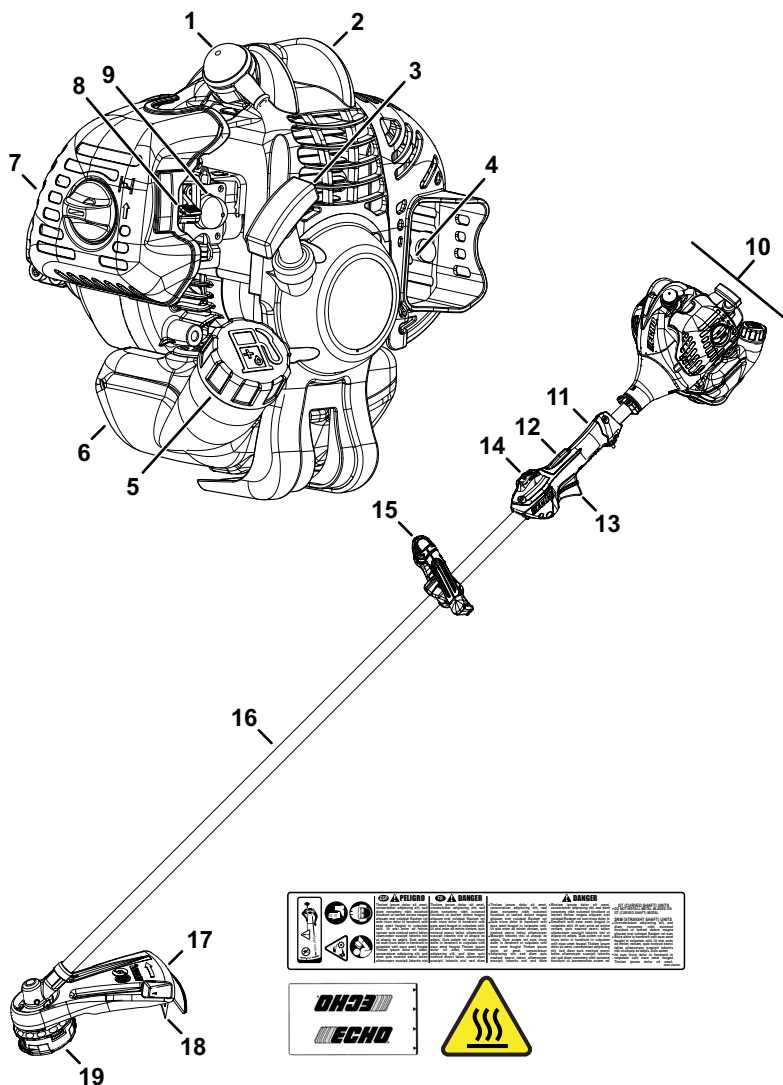
EMISSION CONTROL INFORMATION
ENGINE FAMILY: CEHXS.0214KL DISPLACEMENT: 21.2cc
EMISSION COMPLIANCE PERIOD: 300Hours
THIS ENGINE MEETS 2012 U.S. EPA EXH/EVP & CALIFORNIA
EXH/EVP EMISSION REGULATIONS FOR S.O.R.E. REFER TO OWNER'S
MANUAL FOR MAINTENANCE SPECIFICATIONS AND ADJUSTMENTS.
YAMABIKO CORP. MMM/YYYY 

Product Emission Durability (Emission Compliance Period)

The 50 or 300 hour emission compliance period is the time span selected by the manufacturer certifying the engine emissions output meets applicable emissions regulations, provided that approved maintenance procedures are followed as listed in the Maintenance Section of this manual.

DESCRIPTION

Locate the safety decal(s) on your unit. Make sure the decal(s) is legible and that you understand and follow the instructions on it. If a decal cannot be read, a new one can be ordered from your ECHO dealer. Safety label is for example only. Your label may appear slightly different.



1. Spark Plug
2. Top Guard
3. Recoil Starter Handle
4. Spark Arrester Muffler or Spark Arrester Muffler with Catalyst
5. Fuel Tank Cap
6. Fuel Tank
7. Air Cleaner
8. Choke Lever
9. Purge Bulb
10. Power Head
11. Throttle Handle - For Right Hand
12. Throttle Trigger Lockout
13. Throttle Trigger
14. Stop Switch
15. Support Handle - For Left Hand
16. Drive Shaft Assembly
17. Debris Shield with Cut-Off Knife
18. Cut-Off Knife
19. Nylon Cutter Head

CONTENTS

The ECHO product you purchased has been factory pre-assembled for your convenience. Due to packaging restrictions, some assembly may be necessary.

After opening the carton, check for damage. Immediately notify your retailer or ECHO Dealer of damaged or missing parts. Use the contents list to check for missing parts.

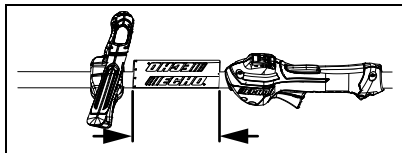
- 1 Power Head / Drive Shaft Assembly
- 1 Quick Start Guide
- 1 Warranty Statement

ASSEMBLY

Support Handle Installation

Note: Label shows minimum spacing for support handle location.

1. If necessary, position support handle for comfortable operation and securely tighten screws.



OPERATION

WARNING

Moving parts can amputate fingers or cause severe injuries. Keep hands, clothing and loose objects away from all openings. Always stop engine, disconnect spark plug, and make sure all moving parts have come to a complete stop before removing obstructions, clearing debris, or servicing unit.

WARNING

Engine exhaust IS HOT, and contains Carbon Monoxide (CO), a poison gas. Breathing CO can cause unconsciousness, serious injury, or death. Exhaust can cause serious burns. ALWAYS position unit so that exhaust is directed away from your face and body.

WARNING

Operation of this equipment may create sparks that can start fires around dry vegetation. This unit is equipped with a spark arrester to prevent discharge of hot particles from the engine. Metal cutters can also create sparks if the cutter strikes rocks, metal, or other hard objects. Contact local fire authorities for laws or regulations regarding fire prevention requirements.

Operation with Blades

WARNING

Metal blades are very sharp and can cause severe injuries, even if unit is off and blades are not moving. Avoid contact with blades. Wear gloves to protect hands.

WARNING

Blade use demands specific brushcutter configuration. Operation without specified shield, barrier bar or U-handle, and harness can result in serious personal injury. Follow installation instructions.

Pro Maxi-Cut Grass / Weed Blade	Tri-Cut Grass / Weed Blade	Metal Grass / Weed Blade	Metal Brush / Clearing Blades
Support Handle, with or without barrier bar	U-handle or support handle with barrier bar		U-handle*
Shield with cut- off knife	Shield without cut-off knife		
Harness	Harness		
Upper plate/flat washer	Upper / Lower blade plates		
Hex nut	Hex nut		
New cotter pin	New cotter pin		

*ANSI standards require brushcutters be equipped with a barrier bar or restrictive harness. U-handle ensures a higher safety factor.

WARNING

Do Not install blades on GT (Curved Shaft) model trimmers.

- Use only ECHO approved parts. Failure to use the correct parts can cause the blade to fly off. Serious injury to the operator and/or bystanders can occur.
- Arbor diameter of upper blade plate must match arbor diameter of blades.

- For barrier bar or U-handle, follow instructions supplied with either blade conversion kit or U-handle kit, and verify blade is secured properly.
- A new cotter pin is required each time a blade is installed.
- Shoulder harnesses may be used on all trimmers and brushcutters to reduce operator fatigue. Brushcutters over 7.5 kg (16.5 lbs.) dry weight (without fuel) and U-handle brushcutters require a double shoulder harness.

Note: The barrier bar is used to restrict rearward movement of the unit. The barrier bar is not a handle and should not be gripped when using or carrying the unit.

Blade Selection

NOTICE

Not all blades are compatible with all trimmers. Visit www.echo-usa.com or www.shindaiwa-usa.com to find compatible blades



WARNING

The type of blade used MUST be matched to the type and size of material cut. An improper or dull blade can cause serious personal injury. Blades MUST be sharp. Dull blades increase the chance of kick-out and injury to yourself and bystanders. Never use an edging blade, circular saw blade, or any other type of unapproved blade.

3-Tooth Grass/Weed Blades may be used wherever the nylon line head is used. DO NOT use this blade for heavy weeds or brush.

8-Tooth Weed/Grass Blade is designed for grass, garden debris and thick weeds up to 19 mm (0.75 in.) diameter. DO NOT use this blade for brush or heavy woody growth.

80-Tooth Brush Blade is designed for cutting brush and woody growth up to 13 mm (0.5 in.) diameter.

22-Tooth Clearing Blade is designed for dense thickets and saplings up to 64 mm (2.5 in.) diameter.



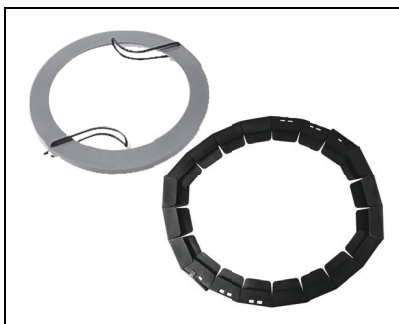
WARNING

A trimmer/brushcutter with a metal blade can cause serious injuries if handled improperly.

Always use extreme care when carrying or handling the equipment to avoid contact with the cutting edges of the blade. Use the optional blade cover when unit is not in use.

Keep blades in protective packaging until ready to install. Store blades safely after removal to prevent injury from accidental contact.

Use blade protectors to protect blade teeth during unit transportation.



Use Shoulder/Waist Harness

Use of the shoulder/waist harness is recommended for all trimmer/brushcutter use, not just blade operation. The shoulder/waist harness when used in a trimming operation with nylon line head suspends the trimmer from the operator's shoulder and reduces operator fatigue.

During blade operation, the same fatigue reduction is achieved. Safety to the operator is also enhanced by reducing the possibility of blade contact with the operator's hands and feet by restricting trimmer movement.



Make sure the warning sign on the back of the shoulder harness can be read easily.

Note: In case of emergency, disconnect the trimmer from the harness.

Fuel

 WARNING

Diesel fuels and alternative fuels, such as E-15 (15% ethanol), E-85 (85% ethanol) or any fuels not meeting ECHO requirements are NOT approved for use in ECHO 2-stroke gasoline engines. Use of diesel or alternative fuels can cause performance problems, loss of power, overheating, fuel vapor lock, and unintended machine operation, including, but not limited to, improper clutch engagement. Diesel or alternative fuels can also cause premature deterioration of fuel lines, gaskets, carburetors and other engine components.

Fuel Requirements

Gasoline - Use fresh (purchased within the last 30 days from the pump) 89 Octane [R+M/2] (mid grade or higher) gasoline known to be good quality. Gasoline may contain up to 10% Ethanol (grain alcohol) or 15% MTBE (methyl tertiary-butyl ether). Gasoline containing methanol (wood alcohol) is **NOT** approved. Use of ECHO branded fuel is recommended to extend engine life in all air-cooled 2-stroke and 2/4-stroke hybrid engines.

Two Stroke Oil - A two-stroke engine oil, such as ECHO branded 2-stroke oils, meeting ISO-L-EGD (ISO/CD 13738) and J.A.S.O. FD Standards must be used. ECHO branded 2-stroke oils meet these standards. Engine problems due to inadequate lubrication caused by failure to use an ISO-L-EGD (ISO/CD 13738) and J.A.S.O. M345-FD certified oil will void the two-stroke engine warranty.

 WARNING

2-Stroke engine oil contains petroleum distillates and other additives that may be harmful if swallowed. Heated oil can release vapors that can cause flash fire, or ignite with explosive force. Read and follow the oil manufacturer's instructions, and observe all safety warnings and precautions for handling flammable liquids. For more detailed safety and first aid information, visit www.echo-usa.com for a copy of the Material Safety Data Sheet.

- ◆ **KEEP OUT OF REACH OF CHILDREN.**
- ◆ **If swallowed, do not induce vomiting. CALL PHYSICIAN OR A POISON CONTROL CENTER IMMEDIATELY.**
- ◆ **WEAR SAFETY GLASSES when mixing or handling.**

- ◆ **AVOID** repeated or prolonged skin contact.
- ◆ **AVOID** inhaling oil mists or vapors.

NOTICE

ECHO branded 2-stroke oils may be mixed at 50:1 ratio for application in all ECHO engines sold in the past regardless of ratio specified in those manuals.

Handling Fuel

DANGER

Fuel is VERY flammable. Use extreme care when mixing, storing or handling, or serious personal injury will result.

- ◆ ***Use an approved fuel container. Mark fuel containers as containing 2-stroke mixture fuel.***
- ◆ ***DO NOT smoke near fuel.***
- ◆ ***DO NOT allow flames or sparks near fuel.***
- ◆ ***Fuel tanks/cans may be under pressure. Always loosen fuel caps slowly allowing pressure to equalize.***
- ◆ ***NEVER refuel a unit when the engine is HOT or RUNNING!***
- ◆ ***DO NOT fill fuel tanks indoors. ALWAYS fill fuel tanks outdoors over bare ground.***
- ◆ ***DO NOT overfill fuel tank. Wipe up spills immediately.***
- ◆ ***Securely tighten fuel tank cap and close fuel container after refueling.***
- ◆ ***Inspect for fuel leakage. If fuel leakage is found, do not start or operate unit until leakage is repaired.***
- ◆ ***Move at least 3 m (10 ft.) from refueling location before starting the engine.***

DANGER

Gasoline vapor is heavier than air, and can travel along the ground to nearby sources of ignition such as electrical motors, pilot lights, and hot or running engines. Vapors ignited by an ignition source can flash back to the fuel container, resulting in an explosion, fire, serious or fatal injuries, and extensive property damage.

Mixing Instructions**NOTICE**

Stored fuel ages. Do not mix more fuel than you expect to use in 30 days, 90 days when a fuel stabilizer is added.

1. Fill an approved fuel container with half of the required amount of gasoline.
2. Add the proper amount of 2-stroke oil to gasoline.
3. Close container and shake to mix oil with gasoline.
4. Add remaining gasoline, close fuel container, and remix.

Fuel to Oil Mix – 50:1 Ratio			
US		Metric	
Gas	Oil	Gas	Oil
gal.	fl.oz.	L	cc
1	2.6	5	100
2	5.2	10	200
5	13	25	500

NOTICE

Spilled fuel is a leading cause of hydrocarbon emissions. Some locations may require the use of automatic fuel shut-off containers to reduce fuel spillage.

Storage - Fuel storage laws vary by locality. Contact your local government for the laws affecting your area. As a precaution, store fuel in an approved, airtight container. Store in a well-ventilated, unoccupied building, away from sparks and flames.

- Empty the fuel tank prior to storing the unit. Return unused fuel to an approved fuel storage container.

NOTICE

Stored two-stroke fuel can separate. ALWAYS shake fuel container thoroughly before each use.

NOTICE

Used oil and gasoline, and soiled towels are hazardous waste materials. Disposal laws vary by locality.

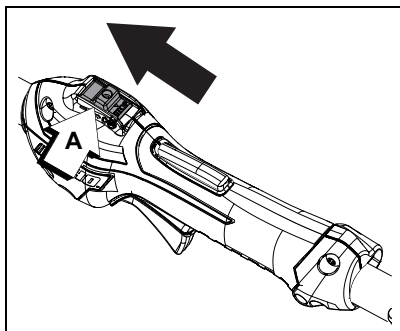
Starting Cold Engine

WARNING

The attachment will operate immediately when the engine starts, and could result in possible serious injury. Keep movable parts of the attachment away from objects that could become entangled or thrown, and surfaces that could cause loss of control.

1. *Stop Switch*

Move stop switch button (A) forward, away from the STOP position.

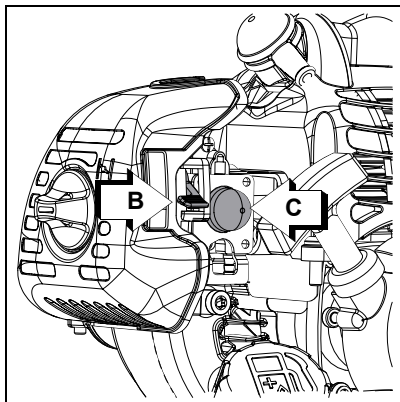


2. *Choke*

Move choke lever (B) to COLD START (|←|) position.

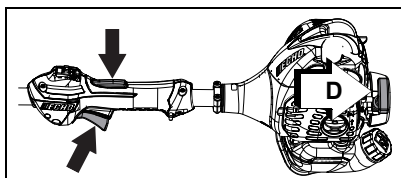
3. *Purge Bulb*

Pump purge bulb (C) until fuel is visible and flows freely in the clear fuel tank return line. Pump bulb an additional four or five times.



4. Recoil Starter

Lay the unit on a flat area and keep movable attachment parts clear of all obstacles. Firmly grasp throttle handle and throttle trigger lockout with left hand and fully depress throttle trigger to wide open position. Rapidly pull recoil starter handle/rope (D) until engine fires (or maximum two to three pulls).



5. Choke

After engine fires (or two to three pulls), move choke (B) to the RUN (|↑|) (open) position. Firmly grasp throttle handle and throttle trigger lockout with left hand and fully depress throttle trigger to wide open position. Pull starter handle/rope until engine starts and runs. Release throttle trigger and allow unit to warm up at idle for several minutes.

Note: If engine does not start with choke in “RUN” position after two to three pulls, repeat instructions 2 - 5.

6. Throttle Trigger

After engine warm-up, grip throttle handle and support handle. Depress the throttle trigger lockout, and gradually depress throttle trigger to increase engine RPM to operating speed.

Starting Warm Engine

The starting procedure is the same as Cold Start except DO NOT close the choke, and do not hold throttle trigger fully depressed.



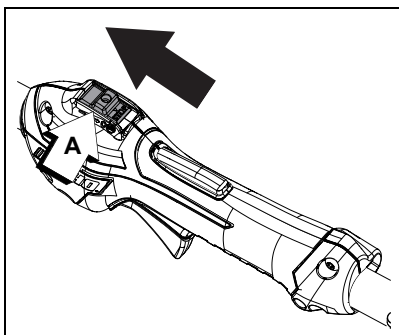
WARNING

The attachment should not move at idle, otherwise serious personal injury may result.

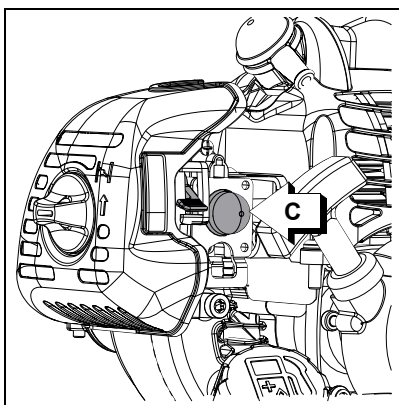
Note: If attachment moves, readjust carburetor according to “Carburetor Adjustment” instructions in this manual or see your ECHO Dealer.

1. Stop Switch

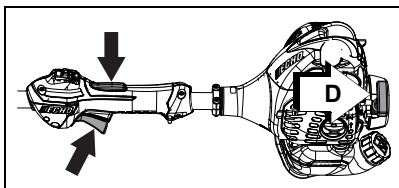
Move stop switch button (A) forward away from the STOP position.

**2. Purge Bulb**

Pump purge bulb (C) until fuel is visible in the "Clear" fuel return line. Pump bulb an additional four or five times.

**3. Recoil Starter**

Lay the unit on a flat area and keep movable attachment parts clear of all obstacles. Firmly grip throttle handle and throttle trigger lockout with left hand. Rapidly pull recoil starter handle/rope (D) until engine fires.



Note: If engine does not start after two to three pulls, use Cold Start Procedure.

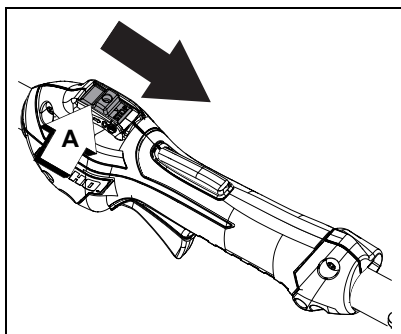
Stopping Engine

1. Throttle

Release throttle trigger and allow engine to return to idle before shutting off engine.

2. Stop Switch

Move stop switch button (A) backward to STOP position.



WARNING

If engine does not stop when stop switch is moved to STOP position, close choke - COLD START position - to stall engine. Have your ECHO dealer repair stop switch before using unit again.

Applications

WARNING

Always hold throttle handle with right hand and support handle with left hand, placing unit on right side of body.

WARNING

Do not install blades on GT (Curved Shaft) trimmers.

Operating Techniques - Nylon Line Head

Nylon line heads may be used for trimming, scything, edging, and scalping of grass and light weeds.

Trimming

Feed the spinning line into the material to be cut. Tilt the line head to one side to direct cutting debris away from you:

- Model SRM/DSRM/PAS/DPAS/SB/TX/C/T (Straight shaft, counter-clockwise line head rotation) - Tilt the cutting head down on the right side while cutting to direct cutting debris away from operator. Feed the line gradually into the material you wish to cut, avoiding contact with fences or other barriers.

- GT models: Tilt cutting head to the left while cutting to direct debris away from the operator.

Scything

Scything - Swing the cutting head in a level arc, gradually feeding the line into the material being cut. Move forward with each arc to cut a swath. Width of cutting swath depends on arc. Use a larger arc for a wider swath, or a smaller arc for a narrow swath. Keep line head tilted to direct cutting debris forward or away from you.

Edging and Scalping

Both of these are done with the nylon line cutting head tilted at a steep angle. Scalping is removing top growth, leaving the earth bare. Edging is trimming the grass back where it has spread over a pavement or driveway. During both edging and scalping, hold the unit at a steep angle in a position where the debris, and any dislodged dirt and stone, will not come back towards you even if it ricochets off the hard surface.

General

- Debris flows in direction of line head rotation. Change line head position to assure debris flow is directed away from operator.
- Keep cutting line away from wire fences to avoid entanglement.
- Operate trimmer only with cutting head below knee height.

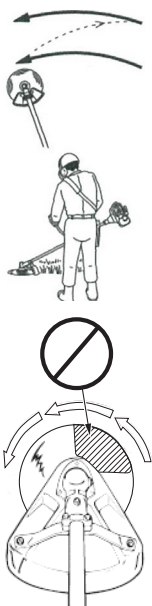
Operating Techniques - Metal or Plastic Blade

Brushcutter blades may be used to cut and trim a wide variety of materials. Refer to the blade selection section for determining the correct blade for the application.

Scything (3, 8, and 80 tooth weed/grass, and brush blades)

- To cut large sections of field grass and weeds swing the cutting head in a level arc, gradually feeding the blade into the material being cut. Adjust throttle speed according to your work.

- Do not swing the main pipe with arms. Turn hips to swing the blade horizontally from right to left, and cut weeds on the left side of the blade.
- Do not scythe back and forth as the grass may scatter and kickback may occur easily.
- Tilt blade left by 5 to 10 degrees so that cut grasses will push left, making progress easier.
- Move forward with each arc to cut a swath.
- Width of cutting swath depends on arc. Use a larger arc for a wider swath, or a smaller arc for a narrow swath. Suggested cutting width is about 1.5 m (4.9 ft).
- When scything large brush up to 12.7 mm (0.5 in.) diameter from right to left, avoid cutting with highlighted section.



Reaction Forces

WARNING

- ◆ The cutting attachment will continue to rotate even after the throttle is released, maintain control of the unit until it has come to a complete stop.
- ◆ Blade thrust may occur when the spinning blade contacts an object that it does not immediately cut. Following proper cutting techniques will prevent blade thrust.
- ◆ Blade thrust can be violent enough to cause the unit and/or operator to be propelled in any direction, and possibly lose control of the unit.
- ◆ Blade thrust can occur without warning if the blade snags, stalls or binds.
- ◆ Blade thrust is more likely to occur in areas where it is difficult to see the material being cut.

Push or Pull - Kickout

During normal use, operating a brushcutter with a circular metal blade can produce sudden strong reaction forces that are difficult to control. Strong reaction forces can cause a loss of balance or loss of control of the equipment, resulting in serious injury to operator and bystanders.

Understanding what causes these reactive forces may help you to avoid them, and can help you to maintain control of the equipment if you experience a sudden reaction during cutting. Reactive forces occur when the force being applied by the cutting teeth of a blade meet resistance, and some of the cutting force is directed back toward the equipment. The greater the cutting force or the amount of resistance, the greater the reactive force.

Push and Pull Forces

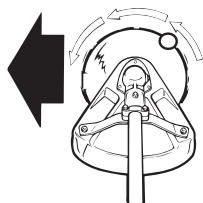
Push and pull forces are reactive forces that push the equipment directly toward the operator, or that pull the equipment directly away from the operator. These forces are the result of cutting on the sides of the blade. The direction of the force depends on the side of the blade being used, and the direction of blade rotation at the point of contact. The reactive force is in the opposite direction of blade rotation at the contact point, regardless of where the contact is being made. These types of reactive forces are also called "Blade Thrust."

As shown in the illustration, a blade turning counterclockwise will cause the equipment to pull away from the operator if the point of cutting resistance is on the left side of the blade. If the point of cutting resistance is on the right side of the blade, the equipment will push back toward the operator. In both examples, the reactive force is in the opposite direction of blade rotation at the contact point where resistance occurs.

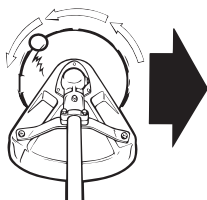
Kickout

Kickout is also a reactive force caused by resistance to cutting, but the direction of blade thrust is lateral (to the left or right of the blade), instead of forward or back toward the operator. In most cases, Push, Pull, and Kickout can be reduced or eliminated by:

- Using the correct blade for the cutting job
- Using properly sharpened blades



- Applying consistent, even force to the blade during the cut
- Avoiding obstacles and ground hazards
- Using extra care when cutting harder materials such as extremely dry brush, saplings, and small trees
- Cutting from a stable, secure position



Blade Cutting Problems

Binding - Blades may bind in the cut if dull or forced. Binding can damage blade, and result in blade breakage or injury from fragments and flying debris. If a blade binds in a cut, don't try to get it out by applying "up and down" force to pry the cut open. Applying prying force to the blade can bend the blade, and result in blade failure and injury.



To free a blade that is bound in the cut, stop the unit, and support the trimmer or brushcutter to keep stress off the blade. Push the tree away from the entry point of the cut to open the cut, and pull the blade directly away from the cut in a straight-line motion. Use caution when releasing the tree to avoid being struck by spring-back or falling.

Inspect the blade for damage before proceeding. Sharpen teeth if dull, or replace blade if cracked, bent, missing teeth, or otherwise damaged.

To prevent binding:

- Keep blades sharp
- Avoid excessive pressure during cuts
- Don't exceed cutting capacity of blade
- Don't use blades with damaged or missing cutting teeth
- Don't rock blades in cut

MAINTENANCE

WARNING

Moving parts can amputate fingers or cause severe injuries. Keep hands, clothing and loose objects away from all openings. Always stop engine, disconnect spark plug, and make sure all moving parts have come to a complete stop before removing obstructions, clearing debris, or servicing unit. Allow unit to cool before performing service. Wear gloves to protect hands from sharp edges and hot surfaces.

WARNING

Operating a poorly maintained unit can result in serious injuries to operator or bystanders. Always follow all maintenance instructions as written, otherwise serious personal injury can result.

Your unit is designed to provide many hours of trouble free service. Regular scheduled maintenance will help your unit achieve that goal. If you are unsure or are not equipped with the necessary tools, we recommend that you take your unit to a Servicing Dealer for maintenance. To help you decide whether you want to DO-IT-YOURSELF or have the Dealer do it, each maintenance task has been graded. If the task is not listed, see your Dealer for repairs.

NOTICE

The use of emission control components other than those specifically designed for this unit is a violation of federal law.

Skill Levels

Level 1 = Easy to do. Common tools may be required.

Level 2 = Moderate difficulty. Some specialized tools may be required.

Level 3 = See your dealer.

Click [HERE](http://www.echo-usa.com/products/maintenance-kit) or go to <http://www.echo-usa.com/products/maintenance-kit>

or

[HERE](http://www.shindaiwa-usa.com/you-can.aspx) <http://www.shindaiwa-usa.com/you-can.aspx>

Maintenance Intervals

COMPONENT / SYSTEM	MAINTENANCE PROCEDURE	SKILL LEVEL
Daily or Before Use		
Air Filter	Inspect / Clean*	1
Choke Shutter		
Fuel System	Inspect ³	
Cooling System	Inspect / Clean	2
Recoil Starter Rope	Inspect / Clean*	1
Screws / Nuts / Bolts	Inspect / Tighten / Replace*	
Blade	Inspect / Replace*	1
Every Refuel		
Fuel System	Inspect ³	1
3 Months		
Air Filter	Replace*	1
Fuel Filter	Inspect*	
Fuel Cap Gasket		
Spark Plug	Inspect / Clean / Replace*	2
Muffler Spark Arrester		
Cylinder Exhaust Port	Inspect / Clean / De-carbon	
Drive Shaft (flex cable models)	Grease ¹	
Gear Housing (some models)	Grease ²	
Yearly		
Fuel Filter	Inspect / Replace*	1
Fuel Cap Gasket	Replace*	

IMPORTANT NOTE - Time intervals shown are maximum. Actual use and your experience will determine the frequency of required maintenance.

MAINTENANCE PROCEDURE NOTES:

¹ Apply lithium-based grease every 25 hours of use.

² Apply lithium-based grease every 50 hours of use.

³ Low evaporative fuel tanks DO NOT require regular maintenance to maintain emission integrity.

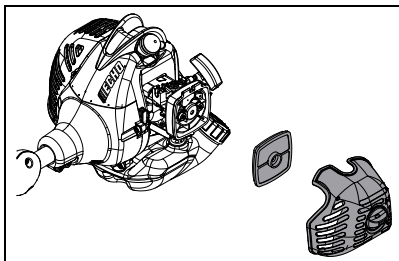
* Replacement is recommended based on the finding of damage or wear during inspection.

Air Filter

Level 1.

Parts required: Tune-Up Kit.

1. Close choke (COLD START position). This prevents dirt from entering the carburetor throat when the air filter is removed. Brush accumulated dirt from air cleaner area.
2. Remove air filter cover. Brush dirt from inside cover.
3. Replace filter if it is damaged, fuel soaked, very dirty, or the rubber sealing edges are deformed.
 - Lightly brush debris from filter.
 - Soak heavily soiled filters in water/detergent solution to loosen dirt, then brush lightly.
 - Rinse with clean water and allow to dry completely before reuse.
4. Install air filter in air filter case.
5. Install air filter cover.



Fuel Filter

Level 1.

Parts required: Tune-Up Kit.

DANGER

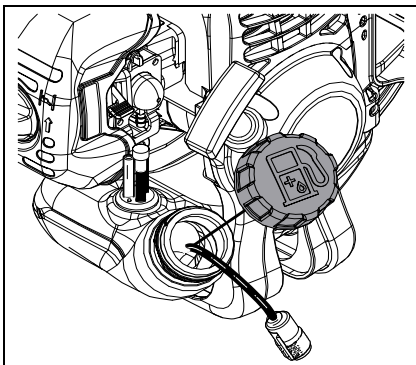
Fuel is VERY flammable. Use extreme care when mixing, storing or handling, or serious personal injury may result.

1. Use a clean rag to remove loose dirt from around fuel cap and empty fuel tank.

NOTICE

Do not damage fuel line while removing fuel filter from tank or line.

2. Pull the fuel filter from the tank.
3. Do not remove the wire coil clamp to remove the fuel filter. Pinch the fuel filter with the fingers on one hand and the fuel line with the other hand. Pull and twist slightly to separate.
4. Install the new filter with the reverse action.



Note: Federal EPA regulations require all model year 2012 and later gasoline powered engines produced for sale in the United States to be equipped with a special low permeation fuel supply hose between the carburetor and fuel tank. When servicing model year 2012 and later equipment, only fuel supply hoses certified by EPA can be used to replace the original equipment supply hose. Fines up to \$37,500 may be enforced for using a non-certified replacement part.

Spark Plug

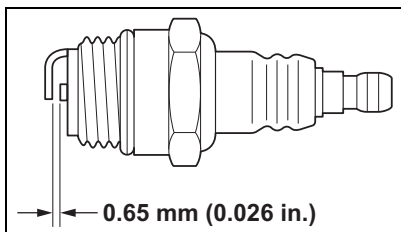
Level 2.

Parts required: Tune-Up Kit.

NOTICE

Use only NGK BPM8Y spark plug (BPMR8Y in Canada) otherwise severe engine damage may occur.

1. Remove spark plug and check for fouling, worn and rounded center electrode.
2. Clean the plug or replace with a new one. DO NOT sand blast to clean. Remaining sand will damage engine.
3. Adjust spark plug gap by bending outer electrode.



4. Tighten spark plug to 130-170 kgf•cm (112-150 lbf•in).

Cooling System

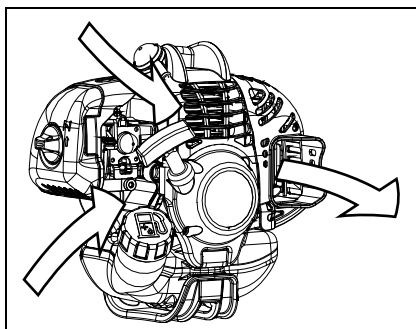
Level 2.

NOTICE

To maintain proper engine operating temperatures, cooling air must pass freely through the cylinder fin area. This flow of air carries combustion heat away from the engine.

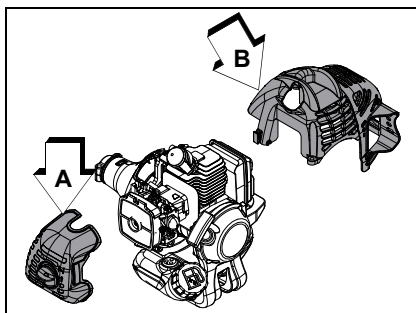
Overheating and engine seizure can occur when:

- Air intakes are blocked, preventing cooling air from reaching the cylinder.
- Dust and grass build up on the outside of the cylinder. This build up insulates the engine and prevents the heat from leaving.



Removal of cooling passage blockages or cleaning of cooling fins is considered "Normal Maintenance." Any failure attributed to lack of maintenance is not warranted.

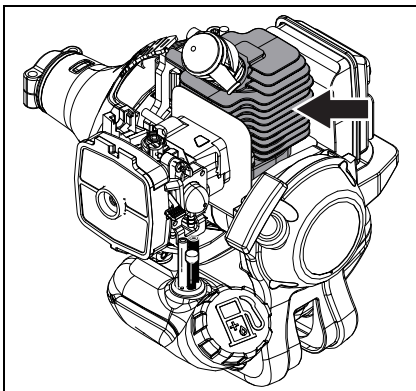
1. Remove spark plug lead.
2. Close choke (COLD START [↔] position). This prevents dirt from entering the carburetor throat when the air filter is removed. Remove air cleaner cover (A).
3. Remove engine cover (B).



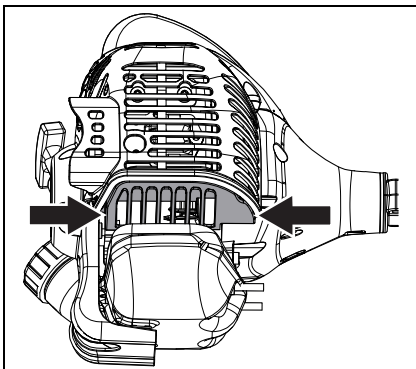
NOTICE

DO NOT use a metal scraper to remove dirt from the cylinder fins.

4. Use a brush to remove dirt from the cylinder fins.
5. Remove ignition wires from clip for cleaning.



6. Remove grass and leaves from the grid between the recoil starter and fuel tank.
7. Assemble components in reverse order.



Exhaust System

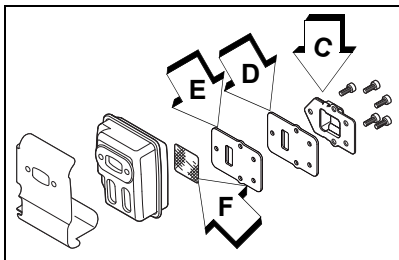
Spark Arrester Screen

Level 2.

Parts required: Spark Arrester Screen, Gasket.

1. Remove spark plug lead and engine cover.
2. Place piston at Top Dead Center (TDC) to prevent carbon/dirt from entering cylinder.
3. Remove spark arrester screen cover (C), gaskets (D), (E), and screen (F), from muffler body.
4. Clean carbon deposits from muffler components.

Note: When cleaning carbon deposit, be careful not to damage the catalytic element inside muffler.



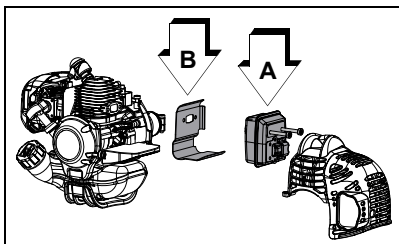
5. Replace screen if it is cracked, plugged, or has holes burned through.
6. Assemble components in reverse order.

Exhaust Port Cleaning

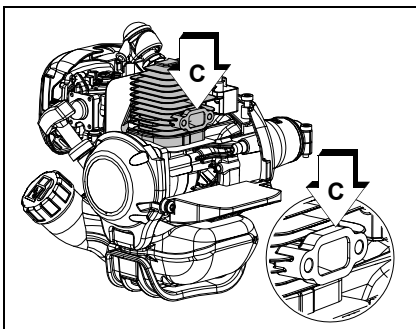
Level 2.

Parts required: Heat Shield (as needed).

1. Remove spark plug lead from spark plug, and remove engine cover.
2. Place piston at top dead center. Remove muffler (A) and heat shield (B).



3. Use a wood or plastic scraping tool to clean deposits from cylinder exhaust port (C).



NOTICE

Never use a metal tool to scrape carbon from the exhaust port. Do not scratch the cylinder or piston when cleaning the exhaust port. Do not allow carbon particles to enter the cylinder.

4. Inspect heat shield, and replace if damaged.
5. Install heat shield and muffler.
6. Tighten muffler mounting bolts (or nuts) to 90-110 kgf•cm (80-95 lbf•in).
7. Install engine cover and attach spark plug lead.
8. Start engine, and warm to operating temperature.
9. Stop engine, and re-tighten mounting bolts (or nuts) to specifications.

Carburetor Adjustment

Level 2.

Engine Break-In

New engines must be operated a minimum duration of two tanks of fuel break-in before carburetor adjustments can be made. During the break-in period your engine performance will increase and exhaust emissions will stabilize. Idle speed can be adjusted as required.

High Altitude Operation

This engine has been factory adjusted to maintain satisfactory starting and durability performance up to 335 m (1,100 ft.) above sea level (ASL) (96.0 kPa). To maintain proper engine operation and emission compliance above 335 m (1,100 ft.) ASL the carburetor may need to be adjusted by an authorized service dealer.

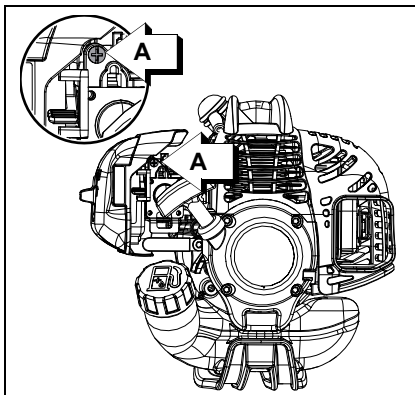
NOTICE

If the engine is adjusted for operation above 335 m (1,100 ft.) ASL, the carburetor must be re-adjusted when operating the engine below 335 m (1,100 ft.) ASL, otherwise severe engine damage may result.

Note: Every unit is run at the factory and the carburetor is set in compliance with emission regulations. Carburetor adjustments, other than idle speed, must be performed by an authorized dealer.

WARNING

- ◆ The cutting attachment may be moving during carburetor adjustments.
 - ◆ Wear your protective equipment and observe all safety instructions.
 - ◆ For units equipped with a clutch, be sure the cutting attachment stops moving when the engine idles.
 - ◆ When the unit is turned off, make sure the cutting attachment has stopped before the unit is set down.
1. Check idle speed and reset if necessary. If a tachometer is available, idle speed screw (A) should be set to the specifications found on "Specifications" page of this manual. Turn idle screw (A) clockwise to increase idle speed; counterclockwise to decrease idle speed.



WARNING

When carburetor adjustment is completed, the cutting attachment should not move at idle, otherwise serious personal injury may result.

Lubrication

Level 1.

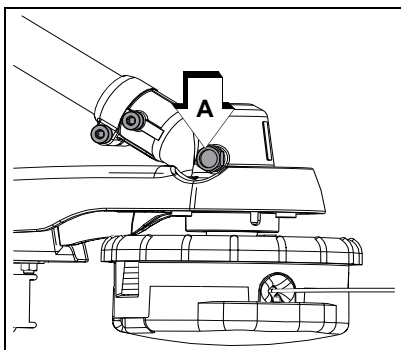
Parts required: Lithium-based grease.

Gear Case

NOTICE

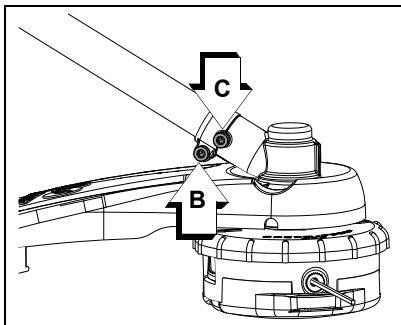
Gear cases without grease plug (A) do not require lubrication.

1. Clean all loose debris from gear case.
2. Remove plug (A) and check level of grease.
3. Add grease if necessary. DO NOT over-fill.

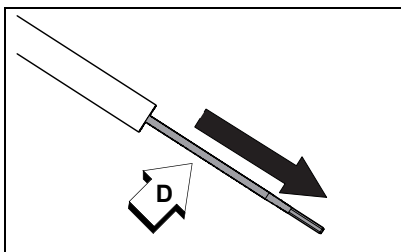


Drive Shaft (flex cable only)

1. Loosen screw (B) and remove locating screw (C). Pull gear case and shield from drive shaft housing.



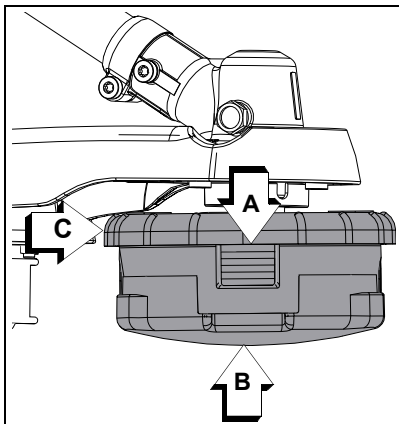
2. Pull flexible cable (D) from the drive shaft housing, wipe clean and re-coat with 15 ml (0.5 oz.) of grease.
3. Slide the flexible cable (D) back in the drive housing. DO NOT get dirt on the flex cable.
4. Install the gear housing and shield assembly.



Nylon Line Head Disassembly Instructions

Note: For normal use, Speed Feed® head disassembly is not necessary. However, if circumstances require disassembly, follow these instructions.

1. Press top of locking tabs (A) on both sides of Speed Feed® head to release cover (B) from knob (C).
2. Remove cover from knob.



Nylon Line Replacement

CAUTION

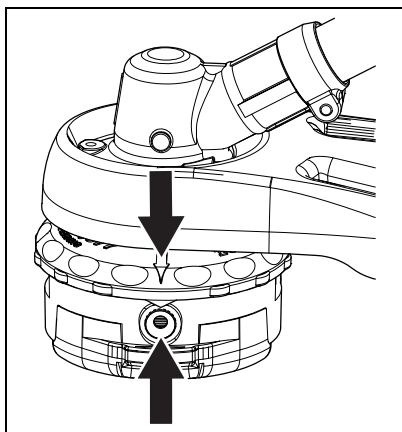
Wear gloves or personal injury may result:

- ◆ *Cut-off knife is sharp.*
- ◆ *Gearcase and surrounding area may be hot.*

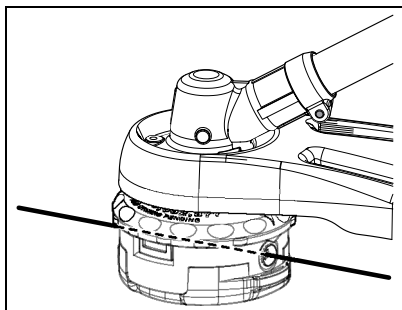
WARNING

Never use wire or wire-rope that can break off and become a dangerous “projectile”. Serious injury can occur.

1. Cut one piece of line to recommended length.
 2.0 mm (0.080 in.) diameter
 6 m (20 ft.).
 2.4 mm (0.095 in.) diameter
 6 m (20 ft.).
2. Align arrows on top of knob with openings in eyelets.

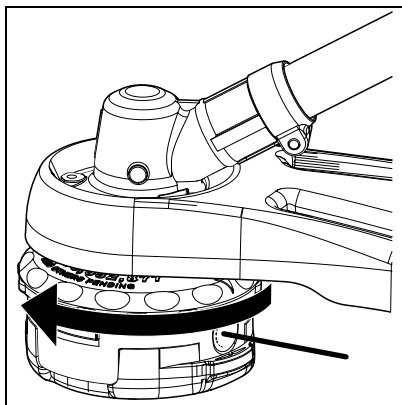


3. Insert one end of trimmer line into an eyelet, and push line equal distance through trimmer head.



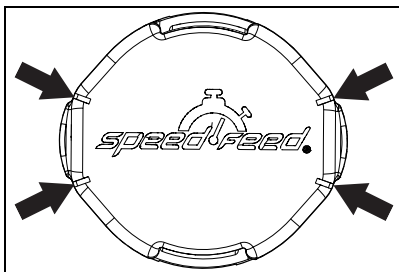
4. Hold trimmer head while turning knob clockwise to wind line onto spool until about 13 cm (5 in.) of each line remains exposed.

Trimmer head is now fully loaded and ready for operation.



NOTICE

When the wear indicators located at the bottom of the Speed-Feed® head are worn smooth, or if holes appear, replacement of the cover or the entire Speed-Feed® head required.



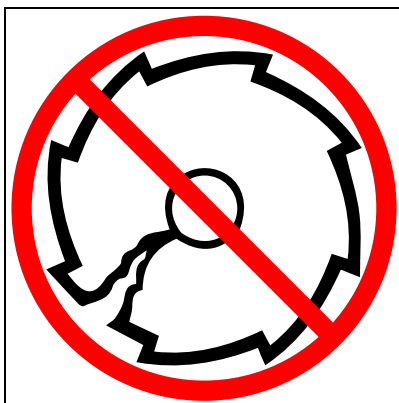
Sharpening Metal Blades

WARNING

Metal blades are very sharp and can cause severe injuries, even if unit is off and blades are not moving. Avoid contact with blades. Wear gloves to protect hands.

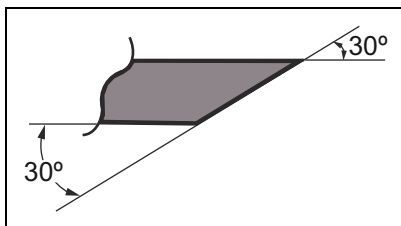
Several styles of metal blades are approved for use on the Brushcutter. The 8-tooth blade can be sharpened during normal maintenance. The clearing blade and 80-tooth blade require professional service.

Before sharpening, CLOSELY inspect blade for cracks (look at the bottom of each tooth and the center mounting hole closely), missing teeth and bending. If ANY of these problems are discovered, replace the blade.



When sharpening a blade, always remove the same amount of materials from each tooth to maintain balance. A blade that is not balanced will cause unsafe handling due to vibration and can result in blade failure.

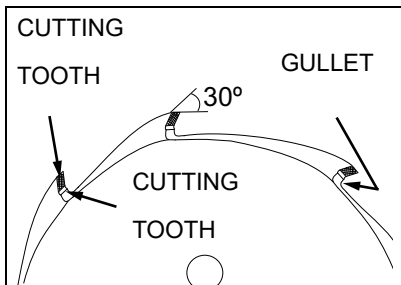
1. File each tooth at a 30° angle a specific number of times, e.g., four strokes per tooth. Work your way around the blade until all teeth are sharp.
2. DO NOT file the 'gullet' (radius) of the tooth with the flat file. The radius must remain. A sharp corner will lead to a crack and blade failure.



NOTICE

If an electric grinder is used, use care not to overheat teeth, do not allow tips/tooth to glow red or turn blue. DO NOT place blade in cooling water. This will change the temper of the blade and could result in blade failure.

3. After sharpening teeth, check each tooth radius for evidence of a square (sharp) corner. Use the round (rat tail) file to renew the radius.



TROUBLESHOOTING

ENGINE PROBLEM TROUBLESHOOTING CHART				
Problem	Check	Status	Cause	Remedy
Engine starts hard or Engine does not start	Fuel at carburetor	No fuel at carburetor	Fuel strainer or fuel line obstructed	Clean or replace See your dealer
	Fuel at cylinder	No fuel at cylinder	Carburetor	See your dealer
		Muffler wet with fuel	Fuel mixture too rich	OPEN choke Clean or replace air filter Adjust carburetor See your dealer
	Spark at end of plug wire	No spark	STOP switch OFF - Electrical problem - Interlock switch	Turn switch to ON See your dealer
	Spark at plug		Incorrect gap - Covered with carbon - Fouled with fuel - Plug defective	Adjust to 0.65 mm (0.026 in.) Clean or replace plug
Engine runs, but dies or Engine does not accelerate properly	Air filter	Air filter dirty	Normal wear	Clean or replace
	Fuel filter	Fuel filter dirty	Contaminants or residue in fuel	Replace filter or replace fuel
	Fuel vent	Fuel vent plugged	Contaminated fuel	Clean or replace

ENGINE PROBLEM TROUBLESHOOTING CHART

Problem	Check	Status	Cause	Remedy
Engine runs, but dies or Engine does not accelerate properly	Spark plug	Plug dirty or worn	Normal wear	Clean and adjust or replace
	Carburetor	Improper adjustment	Vibration	Adjust
	Cooling system	Cooling system dirty or plugged	Extended operation in dirty or dusty locations	Clean
	Spark arrester screen	Spark arrester screen plugged	Normal wear	Replace
Engine does not crank	N/A		Internal engine problem	See your dealer



Fuel vapors are extremely flammable and can cause fire and/or explosion. Never test for ignition spark by grounding spark plug near cylinder plug hole, otherwise serious personal injury can result.

STORAGE

WARNING

During operation the muffler or catalytic muffler and surrounding cover become hot. Always keep exhaust area clear of flammable debris during transportation or when storing, otherwise serious property damage or personal injury may result.

Long Term Storage (Over 30 Days)

Do not store your unit for a prolonged period of time (30 days or longer) without performing protective storage maintenance which includes the following:

1. Store unit in a dry, dust free place, out of the reach of children.

DANGER

Do not store in enclosure where fuel fumes may accumulate or reach an open flame or spark.

2. Place the stop switch in the "OFF" position.
3. Remove accumulation of grease, oil, dirt and debris from exterior of unit.
4. Perform all periodic lubrication and services that are required.
5. Tighten all the screws and nuts.
6. Drain fuel tank completely. Press purge bulb six to seven times to remove remaining fuel from carburetor then drain the tank again. Close choke, start and run the engine until it stops due to lack of fuel.
7. Allow engine to cool. Remove the spark plug lead from the spark plug. Remove the spark plug. Pour 7 cc (0.25 oz.) of fresh, clean, two-stroke engine oil into the cylinder through the spark plug hole.
8. Pull the recoil starter handle two to three times to distribute the oil inside the engine.
9. Observe the piston location through the spark plug hole. Pull the recoil handle slowly until the piston reaches the top of its travel and leave it there.
10. Install the spark plug. Connect the spark plug lead to the spark plug.

SPECIFICATIONS

For units with the following serial numbers:

U00713001001 - U00713999999

U00813001001 - U00813999999

U36513001001 - U36513999999

MODEL	SRM-225
Length (without cutter head)	1,795 mm (70.7 in.)
Width	256 mm (10.1 in.)
Height	329 mm (13 in.)
Weight (dry) (without cutter head)	4.8 kg (10.58 lb.)
Engine Type	Air cooled, two-stroke, single cylinder gasoline engine
Bore	32.2 mm (1.27 in.)
Stroke	26.0 mm (1.02 in.)
Displacement	21.2 cc (1.29 in. ³)
Exhaust	Spark arrester muffler or spark arrester muffler with catalyst
Carburetor	Diaphragm with purge pump
Ignition System	Flywheel magneto, capacitor discharge ignition type
Spark Plug	NGK BPM8Y (BPMR8Y in Canada) Gap 0.65 mm (0.026 in.)
Fuel	Mixed (gasoline and two-stroke oil)
Fuel/Oil Ratio	50:1
Gasoline	Use 89 Octane unleaded. Do not use fuel containing methyl alcohol, more than 10% ethyl alcohol or 15% MTBE. Do not use alternative fuels such as E15 or E85.
Oil	ISO-L-EGD (ISO/CD 13738) and J.A.S.O. M345- FD, two-stroke, air-cooled engine oil.
Fuel Tank Capacity	420 mL (14.2 US fl. oz.)
Starter System	Automatic rewind starter
Clutch	Centrifugal type
Vibration Isolated System	Rubber cushion on engine mount

MODEL	SRM-225
Operating Rod	25.0 mm (1.0 in.) diameter aluminum tube
Drive Shaft	6.35 mm (0.25 in.) flexible shaft
Gear Case Ratio	1.62:1
Rotating Direction	Counterclockwise (viewed from top)
Cutter Head	Speed Feed® 400 LH Nylon line head, Line capacity 6.0 m (20 ft.)
Handle	Front - D-loop type with plastic grip Rear - Plastic grip with throttle trigger and switch
Shoulder Harness	Optional
Idle Speed	3,000 RPM
Clutch Engagement Speed	4,300 RPM
Wide Open Throttle Speed	10,000 RPM

SPECIFICATIONS

For units with the following serial numbers:

U69313001001 - U69313999999

U69513001001 - U69513999999

U72913001001 - U72913999999

MODEL	SRM-225
Length (without cutter head)	1,783 mm (70.2 in.)
Width	256 mm (10.1 in.)
Height	329 mm (13 in.)
Weight (dry) (without cutter head)	4.7 kg (10.4 lb.)
Engine Type	Air -cooled, two-stroke, single -cylinder gasoline engine
Bore	32.2 mm (1.27 in.)
Stroke	26.0 mm (1.02 in.)
Displacement	21.2 cc (1.29 in. ³)
Exhaust	Spark arrester muffler or spark arrester muffler with catalyst
Carburetor	Diaphragm with purge pump
Ignition System	Flywheel magneto, capacitor discharge ignition type
Spark Plug	NGK BPM8Y (BPMR8Y in Canada) Gap 0.65 mm (0.026 in.)
Fuel	Mixed (gasoline and two-stroke oil)
Fuel/Oil Ratio	50:1
Gasoline	Use 89 Octane unleaded. Do not use fuel containing methyl alcohol, more than 10% ethyl alcohol or 15% MTBE. Do not use alternative fuels such as E15 or E85.
Oil	ISO-L-EGD (ISO/CD 13738) and J.A.S.O. M345- FD, two-stroke, air-cooled engine oil.
Fuel Tank Capacity	0.38 L (12.8 US fl. oz.)
Starter System	Automatic rewind starter
Clutch	Centrifugal type
Vibration Isolated System	Rubber cushion on engine mount

MODEL	SRM-225
Operating Rod	25.0 mm (1.0 in.) diameter aluminum tube
Drive Shaft	6.35 mm (0.25 in.) flexible shaft
Gear Case Ratio	1.62:1
Rotating Direction	Counterclockwise (viewed from top)
Cutter Head	Speed Feed® 400 LH Nylon line head, Line capacity 6.0 m (20 ft.)
Handle	Front - D-loop type with plastic grip Rear - Plastic grip with throttle trigger and switch
Shoulder Harness	Optional
Idle Speed	3,000 RPM
Clutch Engagement Speed	4,300 RPM
Wide Open Throttle Speed	10,000 RPM

PRODUCT REGISTRATION

Thank you for choosing ECHO Power Equipment

Please go to <http://www.echo-usa.com/Warranty/Register-Your-ECHO> to register your new product on-line. It's **FAST** and **EASY!** NOTE: your information will never be sold or misused by ECHO, Incorporated.

Registering your purchase enables us to contact you in the unlikely event of a service update or product recall, and verifies your ownership for warranty consideration.

If you do not have access to the Internet, you can complete the form below and mail to:

***ECHO Incorporated, Product Registration, PO Box 1139,
Lake Zurich, IL 60047.***



PRODUCT REGISTRATION CARD : TARJETA DE REGISTRO : CARTE D'ENREGISTREMENT DU PRODUIT

ONLINE OPTION: Register online and save a stamp! You will be alerted of product updates and/or be sent new ECHO product information and special offers. Simply go to:
www.echo-usa.com

OPCIÓN POR INTERNET: Regístrese en la página web y ahorre una estampilla! Le alertarán sobre actualizaciones del producto y/o se le enviará información sobre los nuevos productos ECHO y las ofertas especiales. Simplemente visite:
www.echo-usa.com

OPTION EN LIGNE: Enregistrer en ligne et épargner un timbre! Vous serez informé sur les nouveaux produits ECHO et les mises à jour de produit ainsi que les offres spéciales. Simplement visitez le: www.echo-usa.com

Please Print : Por Favor Escribe Con Letra De Molde : En Lettre Carré S.V.P.

Confidential: Information provided will not be shared or sold

Purchaser's Name : Nombre del Comprador : Nom de l'acheteur	
Address : Dirección : Adresse	
City : Ciudad : Ville	State : Estado : Province
Zip Code : Código Postal : Code Postal	
E-Mail Address : Dirección De Correo Electrónico : Courriel Électronique	
Phone Number : Número De Teléfono : Téléphone	Date of Purchase : Fecha de la Compra : Date de l'achat
Where Purchased : Nombre del Almacén donde fue Comprado : Lieu d'achat	
Model Number : Número del Modelo : No. de modèle	Serial Number : Número de Serie : No. de série

Registering your purchase enables us to contact you in the unlikely event of a service update or product recall and verifies your purchase. We will never sell your information to anyone. Thank you for your purchase.
Como el aviso de tiempo de servicio o la revocación del producto y verificación de propiedad en caso de robo o pérdida.
L'enregistrement de votre achat nous permettra de vous contacter dans le cas improbable de modifications des instructions d'entretien ou de rappel du produit et vérifier que vous êtes le propriétaire en cas de vol ou de perte.

Primary use of tool: ☐ Professional ☐ Homeowner ☐ Rental Equip.

Uso primario de la herramienta: ☐ Profesional ☐ Dueño de casa ☐ Alquiler

Usage principal: ☐ Professionnel ☐ Résidentiel ☐ Location

Do you wish to receive periodic ECHO product information and special offers by e-mail? ☐ Yes ☐ No

¿Desea usted recibir información periódica y ofertas especiales por correo electrónico sobre el producto ECHO? ☐ Yes ☐ No

Souhaitez-vous recevoir périodiquement de l'information sur les produits ECHO et les offres spéciales par courrier électronique? ☐ Oui ☐ Non

Did you visit the ECHO website before purchasing your product? ☐ Yes ☐ No

Visitó usted la página web ECHO antes de comprar su producto? ☐ Si ☐ No

Avez-vous visité le site d'ECHO avant de procurer votre produit? ☐ Oui ☐ Non

What factor(s) most influenced your purchase? ☐ Performance ☐ Price ☐ Dealer ☐ Reliability ☐ Prior Experience ☐ Quality ☐ Friend/Family ☐ Brand ☐ Warranty

¿Qué factor(es) influenció más en su compra? ☐ Precio ☐ Distribuidor ☐ Funcionamiento ☐ Experiencia Anterior ☐ Calidad ☐ Amigo/Familia ☐ Marca ☐ Garantía

Quel facteur (facteurs) vous a influencé le plus? ☐ Performance ☐ Prix ☐ Marchand ☐ Fiabilité ☐ Expérience ☐ Qualité ☐ Ami/Famille ☐ Garantie

ECHO Incorporated 400 Oakwood Rd., Lake Zurich, IL 60047

1.800.432.ECHO WWW.ECHO-USA.COM

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ECHO INCORPORATED

400 Oakwood Road
Lake Zurich, IL 60047
www.echo-usa.com

Trimmer / Brushcutter
Podadora / desmalezadora
Coupe-herbe / débroussailleuse
INTRODUCTION:

This ECHO Quick Start Guide is designed to provide you with details about safety warning symbols, necessary assembly steps, and operational instructions to use your product without delay.

INTRODUCCIÓN:

Esta guía de inicio rápido de ECHO está diseñada para proporcionarle detalles sobre los símbolos de advertencia de seguridad, los pasos necesarios para el ensamblaje y las instrucciones de operación para usar el producto sin demora.

INTRODUCTION:

Ce Guide de démarrage rapide ECHO est conçu pour vous fournir les détails sur les signaux de sécurité et d'avertissement, les étapes de montage nécessaires et les instructions opérationnelles qui vous permettront d'utiliser votre produit dans les plus brefs délais.



Scan QR Code or visit the web link www.echo-usa.com/manuals or call 1-800-432-ECHO (3246) for the Operators Manual, additional literature including safety manual, a copy of the product warranty, or to register your product. Before calling or accessing the website, know the model and serial number of your unit.

Escanee el código QR o visite el enlace al sitio web www.echo-usa.com/manuals o llame al 1-800-432-ECHO (3246) para obtener el Manual del operador, la bibliografía adicional incluido un manual de seguridad, una copia de la garantía del producto, o para registrar el producto. Antes de llamar o acceder al sitio web, conozca el modelo y el número de serie de su unidad.

Scannez le code QR ou visitez le lien www.echo-usa.com/manuals ou appelez au 1 800 432-ECHO (3246) pour avoir le Manuel de l'utilisateur, une documentation complémentaire tels que le manuel de sécurité, une copie de la garantie du produit, ou encore pour enregistrer votre produit. Avant d'appeler ou d'accéder au site Web, connaissez le modèle et le numéro de série de votre appareil.

INTERNATIONAL SAFETY SYMBOLS / SÍMBOLOS DE SEGURIDAD INTERNACIONALES / SYMBOLES INTERNATIONAUX DE SÉCURITÉ

Symbols / Símbolos / Symboles	International Symbols	Símbolos internacionales	Symboles internationaux
	Warning, See Operator's Manual	Lea y entienda el manual del operador	Lire et veiller à bien comprendre les instructions du manuel d'utilisation
	Wear Eye, Ear and Head Protection	Protéjase los ojos, oídos y cabeza	Porter des protections pour les yeux, les oreilles et la tête
	Wear Hand and Foot Protection	Use protection para pies y manos	Usure protection de main et de pied
	Safety / Alert	Seguridad / alerta	Sécurité / alerte
	DO NOT Allow Flames or Sparks Near Fuel	NO permita que haya llamas o chispas cerca del combustible	le carburant de flammes ou d'étincelles
	DO NOT Smoke Near Fuel	NO fume cerca del combustible	NE PAS fumer près du carburant
	Hot Surface	Superficie caliente	Surface brûlante
	Fuel and Oil Mixture	Mezcla de combustible y aceite	Mélange huile et essence
	Choke Control "RUN" Position (Choke Open)	Control del estrangulador "marcha" posición (estrangulador abierto)	Starter volet de départ position RUN (ouvert)
	Choke Control "COLD START" Position (Choke Closed)	Control del estrangulador "arranque en frío" posición (estrangulador cierre)	Starter volet de départ position START (fermé)
	Keep Feet Away From Blade	No acerque los pies a la hoja	Garder les pieds à l'écart de la lame
	Thrown Objects	Objetos lanzados	Projection d'objets
	Rotating Cutting Attachment	Accesorio giratorio del corte	Accessoire de rotation de la coupe
	Direction of Blade	Dirección de la hoja	Direction de la lame
	Keep Bystanders and Helpers Away 15 m (50 ft.)	No permita a nadie que se acerque más de 15 metros (50 pies) de la máquina	Ne laisser personne approcher à moins de 15 m (50 pi)

MANUAL SAFETY SYMBOLS AND IMPORTANT INFORMATION

Throughout this manual and on the product itself, you will find safety alerts and helpful, informational messages preceded by symbols or key words. The following is an explanation of those symbols and key words and what they mean to you.

DANGER

The safety alert symbol accompanied by the word **"DANGER"** calls attention to an act or condition which **WILL** lead to serious personal injury or death if not avoided.

WARNING

The safety alert symbol accompanied by the word **"WARNING"** calls attention to an act or condition which **CAN** lead to serious personal injury or death if not avoided.

NOTICE

The enclosed message provides information necessary for the protection of the unit.

Note: This enclosed message provides tips for use, care and maintenance of the unit.

GENERAL SAFETY REQUIREMENTS

Read and understand all provided literature before use. Failure to do so could result in serious injury. Additional operating instructions are available from your Authorized ECHO Dealer.

WARNING

- Operating a poorly maintained unit can result in serious injuries to operator or bystanders. Always follow all maintenance instructions as written, otherwise serious personal injury can result.
- Use only approved attachments. Serious injury can result from the use of a non-approved attachment combination.
- Do not attempt to modify this product. Serious injury can result from the use of any modified product.
- Do not operate this unit when tired, ill, or under the influence of alcohol, drugs, or medication. Serious injury can result from the use of this product in an impaired state.
- Moving parts can amputate fingers or cause severe injuries. Keep hands, clothing and loose objects away from all openings. Always stop engine, disconnect spark plug, and make sure all moving parts have come to a complete stop before removing obstructions, clearing debris, or servicing unit.

Use The Correct Personal Protection

WARNING

ALWAYS WEAR	NEVER WEAR
Hearing Protection	Loose Clothing
Eye Protection	Jewelry
Heavy, Long Pants	Short Pants, Short Sleeve Shirt
Boots	Sandals
Gloves	Barefoot
Long-Sleeve Shirt	Long Hair Below Shoulders

Before Each Use Inspect:

- For damaged parts.
- Loose or missing fasteners.
- Cutting attachments for damage (cracked, chipped, etc.).
- Cutting attachment is securely fastened.
- Cutting shield is correct for cutting attachment and secured according to this manual.
- For fuel leaks from any point of the fuel system (tank to carburetor).
- Manufacturer recommended flexible non-metallic line is installed in the trimmer head.

Proper Operating Position

WARNING

- The use of the optional shoulder / waist harness for ALL use of the mower / cutter is recommended, not only for the operation of the blade.
- Keep a firm grip on both handles.
- Be sure front handle is installed according to assembly instructions.
- For barrier bar or u-handle, follow instructions supplied with either blade conversion kit or U-handle kit.
- Keep firm footing and balance.
- Do not over-reach.
- Keep cutting attachment below waist.
- Keep all body parts away from rotating cutting attachment and hot surfaces.



WARNING

If the cutting attachment moves at idle please adjust according to the section in the operator's manual.

Exhaust Gases

WARNING

Do not operate this product indoors or in inadequately ventilated areas. Engine exhaust contains poisonous emissions and can cause serious injury or death.

Fuel Handling

DANGER

- Fuel is VERY flammable. Use extreme care when mixing, storing or handling, or serious personal injury can result.
- Mix and pour fuel outdoors where there are no sparks and flames.
- Slowly remove the fuel cap only after stopping the engine.
- Do not smoke while fueling or mixing fuel.
- Wipe spilled fuel from the unit.
- Move at least 3 m (10 ft) away from the fueling source and site before starting engine.

Work Area

- Remove all objects such as rocks, broken glass, nails, wire, or string, which can be thrown or become entangled in the cutting attachment.
- Clear the area of children, bystanders and pets.
- At a minimum, keep all children, bystanders and pets outside a 15m (50 ft) radius.
- Outside the 15 m (50 ft) zone, there is still a risk of injury from thrown objects.
- Bystanders should be encouraged to wear eye protection.
- If you are approached, stop the engine and cutting attachment.
- When a bladed unit is used, there is the added risk of injury to bystanders being struck with the moving blade in the event of a blade thrust or other unexpected reaction of the blade.

INFORMACIÓN IMPORTANTE Y SÍMBOLOS DE SEGURIDAD DEL MANUAL

A lo largo de este manual y en el producto mismo, encontrará alertas de seguridad y mensajes informativos de gran utilidad precedidos por símbolos o palabras clave. A continuación se enumera una explicación de dichos símbolos y palabras clave y su significado.

PELIGRO

El símbolo de alerta de seguridad acompañado por la palabra "PELIGRO" destaca un acto o condición que LLEVARÁ a que se produzcan lesiones personales graves o la muerte si no se evita.

ADVERTENCIA

El símbolo de alerta de seguridad acompañado por la palabra "ADVERTENCIA" destaca un acto o condición que PUEDE llevar a que se produzcan lesiones personales graves o la muerte si no se evita.

AVISO

El mensaje adjunto proporciona información necesaria para proteger la unidad.

Nota: Este mensaje adjunto proporciona sugerencias para el uso, cuidado y mantenimiento de la unidad.

REQUISITOS DE SEGURIDAD GENERALES

Lea y comprenda toda la bibliografía provista antes del uso. De lo contrario, pueden producirse lesiones graves. Puede obtener instrucciones de operación adicionales en su distribuidor autorizado de ECHO.

ADVERTENCIA

- La operación de una unidad en condiciones de mantenimiento deficientes puede ocasionar lesiones graves en el operador o las demás personas presentes. Siempre siga todas las instrucciones de mantenimiento como están escritas; de lo contrario, pueden producirse lesiones personales graves.
- Use solo los accesorios aprobados. Pueden producirse lesiones graves si se usa una combinación no aprobada de accesorios.
- No intente modificar este producto. Pueden producirse lesiones graves si se usa una modificación de este producto.
- No opere esta unidad cuando esté cansado, enfermo, o si está bajo los efectos del alcohol, las drogas o medicamentos. Pueden producirse lesiones graves si se usa este producto en un estado de deterioro.
- Las piezas móviles pueden amputar dedos o causar lesiones graves. Mantenga las manos, la vestimenta y los objetos sueltos alejados de todas las aberturas. Siempre apague el motor, desconecte la bujía y asegúrese de que todas las partes móviles se hayan detenido por completo antes de eliminar obstrucciones, limpiar suciedad o realizar tareas de reparación en la unidad.

Utilizar La Protección Personal Correcta

ADVERTENCIA

USE SIEMPRE	NO USE NUNCA
Protección auditiva	Ropa suelta
Protección ocular	Joyas
Pantalones largos y gruesos	Pantalones cortos y camisa de mangas cortas
Botas	Sandalias
Guantes	Descalzo
Camisa de mangas largas	Cabello largo por debajo de los hombros

ANTES DE CADA USO, INSPECCIONAR:

- Si hay piezas dañadas.
- Si hay sujetadores sueltos o faltantes.
- Si hay daño en los accesorios de corte (rajados, rotos, etc.).
- Si el accesorio de corte está asegurado firmemente.
- Si la placa de corte es correcta para el accesorio de corte y está asegurada según este manual.
- Si hay pérdidas de combustible de algún punto del sistema de combustible (del tanque al carburador).
- Si la línea no metálica flexible recomendada por el fabricante está instalada en el cabezal de la podadora.

Posición correcta de operación

ADVERTENCIA

- L'utilisation du harnais épaule / ceinture est recommandée pour TOUT usage des débroussailluses et débroussailluses, pas seulement pour les lames.
- Sujete firmemente los dos mangos.
- Asegúrese de que el mango delantero esté instalado según las instrucciones de ensamblaje.
- Para la barra de barrera o el mango en U, siga las instrucciones que se suministran con cada kit de conversión de hoja o kit de mango en U.
- Mantenga una posición estable y un equilibrio firme.
- No se estire.
- Mantenga el accesorio de corte debajo de la cintura.
- Mantenga todas las partes del cuerpo alejadas del accesorio de corte giratorio y de las superficies calientes.



ADVERTENCIA

Si el accesorio de corte se mueve en ralentí, ajústelo según la sección del manual del operador.

Gases del escape

ADVERTENCIA

No haga funcionar este producto en interiores o en áreas mal ventiladas. El sistema de escape del motor contiene emisiones tóxicas que pueden producir lesiones graves o la muerte.

Manipulación del combustible

PELIGRO

- El combustible es MUY inflamable. Tenga extremo cuidado al mezclar, almacenar o manipular el combustible, o pueden producirse lesiones personales graves.
- Mezcle y vierta el combustible en exteriores donde no haya chispas ni llamas.
- Retire lentamente el tapón de combustible solo después de detener el motor.
- No fume mientras abastece o mezcla el combustible.
- Limpie el combustible derramado de la unidad.
- Muévase al menos 3 m (10 ft) de la fuente y el sitio de abastecimiento de combustible antes de arrancar el motor.

Área de trabajo

- Quite todos los objetos como piedras, vidrios rotos, clavos, cables o cuerdas que puedan lanzarse o quedar atascados en el accesorio de corte.
- Despeje el área de niños, otras personas presentes y mascotas.
- Como mínimo, mantenga a todos los niños, otras personas presentes y mascotas fuera de un radio de 15 m (50 ft).

- Fuera de la zona de 15 m (50 ft), aún hay riesgo de sufrir lesiones a causa de los objetos lanzados.
- Se debe alertar a las personas presentes a usar protección ocular.
- Si alguien se acerca a usted, detenga el motor y el accesorio de corte.
- Cuando se usa una unidad de hojas, existe el riesgo adicional de que las personas presentes sufran lesiones si los golpea la hoja en movimiento en el caso de un contragolpe de la hoja u otra reacción inesperada de la hoja.

SYMBOLES DE SÉCURITÉ DANS LE MANUEL ET INFORMATIONS IMPORTANTES

Tout au long de ce manuel et sur le produit lui-même, vous trouverez des alertes de sécurité et des messages d'information, précédés par des symboles ou des mots clés. Voici une explication des symboles et des mots clés et ce qu'ils signifient pour vous.

DANGER

Les symboles d'alerte de sécurité accompagné du mot « **DANGER** » attirent l'attention d'un acte ou condition qui **MÈNERA** à des blessures graves ou décès, si non évité.

AVERTISSEMENT

Les symboles d'alerte de sécurité accompagné du mot « **AVERTISSEMENT** » attirent l'attention d'un acte ou condition qui **PEUT** mener à des blessures graves ou décès, si non évité.

AVIS

Le message ci-joint fournit les informations nécessaires à la protection de l'unité.

Remarque: Ce message ci-joint fournit des conseils pour l'utilisation, soins et maintenance de l'unité.

EXIGENCES GÉNÉRALES DE SÉCURITÉ

Lire et comprendre toute la documentation à votre disposition avant utilisation. Le non-respect de ces instructions peut entraîner de graves blessures. Des instructions d'utilisation supplémentaires sont disponibles auprès de votre revendeur ECHO agréé.

ATTENTION

- L'utilisation d'un appareil mal entretenu peut causer de graves blessures à l'utilisateur ou aux curieux. Toujours suivre les instructions d'entretien telles que prescrites, sinon de graves blessures peuvent être causées.
- N'utiliser que des attaches approuvées. L'utilisation de combinaisons d'accessoires non-approuvés peut causer de graves blessures.
- Ne pas essayer de modifier ce produit. L'utilisation d'un produit modifié peut causer de graves blessures.
- Ne pas manipuler ce dispositif lorsque fatigué, malade ou sous l'influence de l'alcool, de drogues ou de médicaments. L'utilisation du produit lorsqu'on est en état affaibli ou affecté peut causer de graves blessures.
- Les pièces mobiles peuvent amputer les doigts et causer de graves blessures. Garder les mains, les vêtements et autres objets mobiles éloignés de toutes les ouvertures. Toujours couper le moteur, débrancher la bougie et s'assurer que les pièces mobiles sont à l'arrêt, avant de retirer les obstructions, de sortir les débris ou de faire réviser l'appareil.

Utilisez La Protection Personnelle Appropriée

ATTENTION

TOUJOURS PORTER	NE JAMAIS PORTER
Protection auditive	Vêtements amples
Protection des yeux	Bijoux
Pantalons épais et longs	Pantalons courts, chemises à manches courtes
Bottes	Sandales
Gants	Pieds nus
Chemises à manches longues	Cheveux dont la longueur dépasse les épaules

INSPECTER AVANT CHAQUE UTILISATION :

- en cas de pièces endommagées.
- en cas d'attaches manquantes ou mal fixées
- en cas de dommages sur les accessoires de coupe (cassé, ébréché, etc.).
- pour vérifier que l'accessoire de coupe est bien fixé.
- pour s'assurer que le panneau tranchant correspond à l'accessoire de coupe et est fixé suivant les prescriptions de ce manuel.
- en cas de fuites d'essence par endroits du système (du réservoir au carburateur).
- pour vérifier que le fil flexible et non métallique recommandé par le fabricant est installé dans le coupe-herbe.

Position De Fonctionnement Adéquat

AVERTISSEMENT

- L'utilisation du harnais optionnel épaule / ceinture est recommandée pour TOUT usage des débroussailleuses / débroussailleuses, pas seulement pour les lames.
- Tenir l'appareil fermement avec les deux mains.
- S'assurer que la poignée avant est installée selon les instructions d'assemblage.
- Pour les barres de protection ou les poignées en U, suivre les instructions fournies dans la trousse de conversion de la lame ou dans la trousse de la poignée en U.
- Bien placer ses pieds et garder l'équilibre.
- Ne trop se pencher pour atteindre
- Garder l'accessoire de coupe en-dessous de la ceinture.
- Tenir toutes les parties du corps éloignés des accessoires rotatifs de coupe et des surfaces chaudes.



AVERTISSEMENT

Si l'accessoire de coupe marque un arrêt, veuillez l'ajuster selon les instruction de la section du manuel de l'utilisateur.

Les gaz d'échappement

AVERTISSEMENT

Ne pas opérer ce produit à l'intérieur ou dans des pièces qui ne sont pas suffisamment aérées. Les gaz d'échappement du moteur sont toxiques et peuvent causer des dommages graves ou même la mort.

Manutention de l'essence

⚠ DANGER

- L'essence est TRÈS inflammable. Faire très attention pendant le mélange, la conservation et la manipulation; sinon des blessures graves peuvent survenir.
- Mélanger et verser l'essence à l'extérieur, loin des flammes et des étincelles.
- Ne retirer lentement le bouchon de réservoir du carburant qu'après avoir arrêté le moteur.
- Ne pas fumer lorsqu'on ravitaille le moteur ou lorsqu'on mélange l'essence.
- Essuyer le carburant déversé sur l'appareil.
- Se tenir à une distance d'au moins 3 m (10 pi) de la source et du site de ravitaillement avant de démarrer le moteur.

Aire de travail

- Retirer tout objet pouvant être rejeté ou obstruer l'accessoire de coupe, comme des pierres, du verre cassé, des clous, des câbles ou des fils.
- Éviter de travailler en présence des enfants, de curieux et d'animaux domestiques.
- Au mieux, garder les enfants, curieux et animaux domestiques en dehors d'un rayon de 15 m (50 pi).
- Il y a encore des risques de blessures en dehors de la zone des 15 m (50 pi) avec les objets rejetés.
- Les curieux doivent porter des lunettes protectrices.
- Arrêter le moteur et l'accessoire de coupe lorsque l'on s'approche de vous.
- Pendant l'utilisation d'un appareil à lame, il y a grand risque de blesser les curieux avec la lame mobile, en cas de ricochet ou de toute autre réaction inattendue de la lame.

HANDLE POSITIONING / POSICIONAMIENTO DEL MANGO / POSITION DES POIGNÉES

1

A SRM

B

B SRM

A

A GT

B

B GT

X Label shows minimum spacing for support handle location.
La etiqueta muestra una separación mínima para la ubicación de empuñadura de soporte.
L'illustration montre Separación empuñadura de soporte para poignée maintien.

FUEL HANDLING / MANIPULACIÓN DEL COMBUSTIBLE / MANIPULATION DU CARBURANT

2

OR

A

50 : 1

B

COLD START / ARRANQUE EN FRÍO / DÉMARRAGE À FROID

3

A

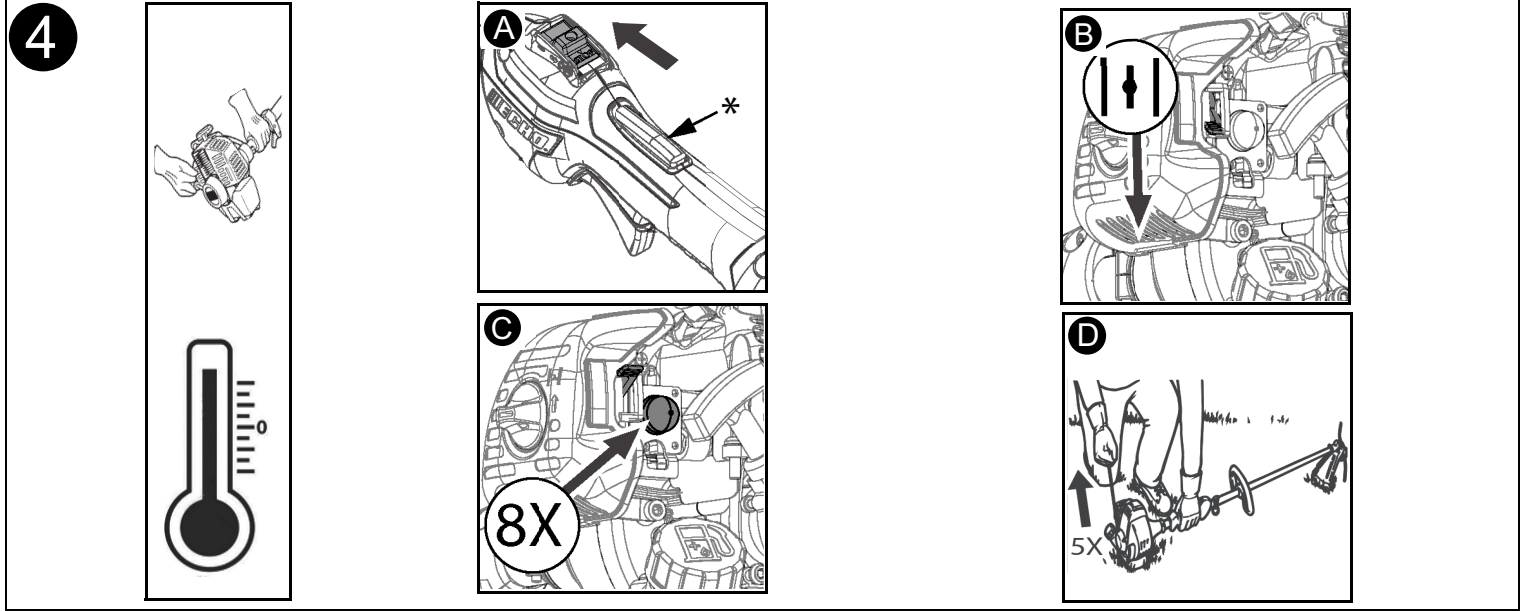
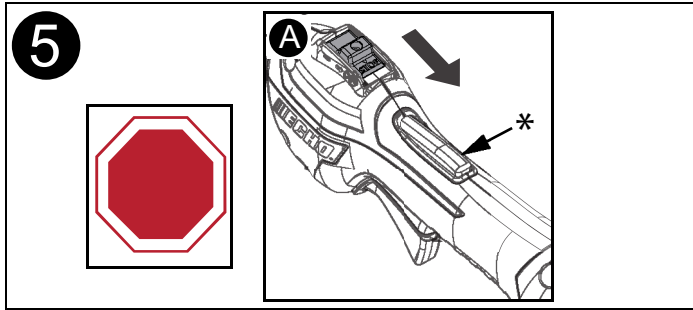
B

C

D

E

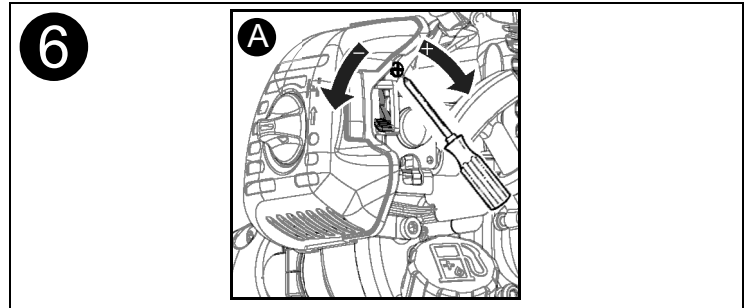
F

WARM START / ARRANQUE EN CALIENTE / DÉMARRAGE À CHAUD

STOP ENGINE / DETENER MOTOR / ARRÊT MOTEUR


* - If equipped with Throttle Trigger Lockout.

* - Si cuenta con bloqueo del gatillo del acelerador.

* - Si équipé avec le verrouillage de la gâchette.

**MAINTENANCE PROCEDURE: IDLE SPEED ADJUSTMENT
PROCEDIMIENTO DE MANTENIMIENTO: AJUSTE DE LA VELOCIDAD DE RALENTÍ**
PROCESSUS D'ENTRETIEN RÉGLAGE DE LA VITESSE DE RALENT

MAINTENANCE INTERVALS / INTERVALOS DE MANTENIMIENTO / INTERVALLES D'ENTRETIEN

<i>Daily Or Before Use / Diariamente O Antes De Usar / Quotidienn Ou Avant Utilisation</i>	
Air Filter, Choke Shutter Filtro de aire, Estrangulador Filtre à air, Volet de départ	Inspect-Clean* Inspeccionar-Limpiar* Inspection-Nettoyage*
Cooling System, Recoil Starter Rope Sistema de enfriamiento, Cuerda del motor de arranque Système de refroidissement, Cordon lanceur	Inspect-Clean* Inspeccionar-Limpiar* Inspection-Nettoyage*
Screws, Nuts, Bolts Tornillos, tuercas, pernos Vis, écrous, boulons	Inspect-Tighten-Replace* Inspeccionar-Apretar-Reemplazar * Inspection-Serrage-Remplacement*
3 Months / 3 Meses / 3 Mois	
Air Filter Filtro de aire Filtre à air	Replace* Reemplazar* Remplacement*
Fuel Filter, Fuel Cap Gasket Filtro de combustible, Empaquetadura de la cubierta del combustible Filtre à carburant, Joint de couverture de carburant	Inspect* Inspeccionar* Inspection*

Spark Plug, Muffler Spark Arrestor Bujía, Apagachispas del silenciador Bougie, Pare-étincelles	Inspect-Tighten-Replace* Inspeccionar-Apretar-Reemplazar* Inspection-Serrage-Remplacement*
Drive Shaft (Flex Cable Models) Eje de impulsión (modelos de cable flexible) Arbre moteur (modèles à arbre flexible).	Apply lithium based grease every 25 hours of use. Aplique el grasa a base de litio grasar cada 25 horas de uso. Appliquer le graisse au lithium toutes les 25 heures a utilisation.
Gear Housing (Some Models) Caja de engranajes (algunos modelos) Carter d'engrenages (certains modèles).	Apply lithium based grease every 50 hours of use. Aplique el grasa a base de litio grasar cada 50 horas de uso. Appliquer le graisse au lithium toutes les 50 heures a utilisation.
Yearly / Anualmente / Annuellement	
Fuel Filter Filtro de combustible Filtre à carburant	Inspect-Replace* Inspeccionar-Reemplazar * Inspection-Remplacement*
IMPORTANT NOTE - Time intervals shown are maximum. Actual use and your experience will determine the frequency of required maintenance. * Replacement is recommended based on the finding of damage or wear during inspection. NOTA IMPORTANTE - Los intervalos indicados son máximos. El uso real y su experiencia determinarán la frecuencia del mantenimiento requerido. * Todas las recomendaciones de reemplazo se basan en encontrar piezas dañadas o desgastadas durante la inspección. REMARQUE IMPORTANTE - Intervalles de temps sont maximales. Utilisation réelle et votre expérience détermineront la fréquence d'entretien requis. * Toutes les recommandations de remplacement de pièces sont basées sur la découverte de dommages ou d'usure lors de l'inspection.	

MAINTENANCE

Spark Arrestor Screen

Parts Required: Spark Arrestor Screen, Gasket

1. Remove spark plug lead and engine cover.
2. Place piston at Top Dead Center (TDC) to prevent carbon/dirt from entering cylinder.
3. Remove spark arrestor screen cover, gaskets, and screen, from muffler body.
4. Clean carbon deposits from muffler components.

Note: When cleaning carbon deposit, be careful, do not damage the catalytic element inside muffler (If equipped with catalytic element).

5. Replace screen if it is cracked, plugged, or has holes burned through.
6. Assemble components in reverse order.

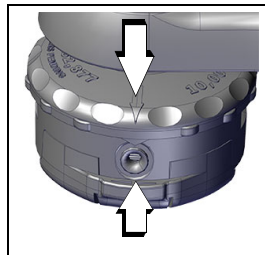
Nylon Line Replacement

WARNING

- Never use wire or wire-rope that can break off and become a dangerous "projectile". Serious injury can occur.
- Wear gloves or personal injury can result:
- Cutoff knife is sharp.
- Gearcase and surrounding area can get hot.

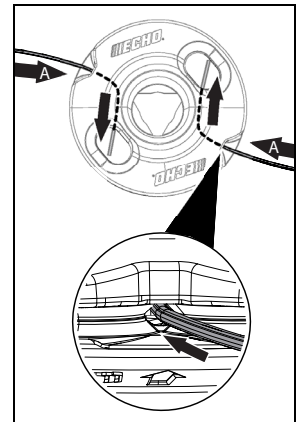
Speed Feed™

1. Cut one piece of 2.0 mm (0.80 in.) or 2.4 mm (0.95 in.) line to recommended length of 6 m (20 Ft.).
2. Align arrows on top of knob with opening of eyelets.
3. Insert one end of trimmer line into eyelet and push line until equal lengths extend from trimmer head.
4. Hold trimmer head and turn knob to wind line onto spool.
5. Continue until approximately 13 cm (5 in.) of line extends on each side.



Rapid Loader™

1. Shut engine OFF. Lay unit on the ground with head assembly up.
2. Insert one piece of trimmer line through eyelet (A) on each side of head. Route line as shown.
3. Remove old nylon line in direction shown.



Idle Speed Adjustment

NOTICE

- Every unit is run at the factory and the carburetor is set in compliance with emission regulations. Carburetor adjustments, other than idle speed, must be performed by an authorized ECHO dealer.
- If a tachometer is available, idle speed screw (A) (Figure 6A) should be set to the specifications found in the Operator's Manual. Turn idle screw (A) clockwise to increase idle speed; counterclockwise to decrease idle speed.

WARNING

- The cutting attachment may be spinning during carburetor adjustments.
- Wear your protective equipment and observe all safety instructions.
- For units equipped with a clutch, be sure the cutting attachment stops turning when the engine idles.
- When the unit is turned off, make sure the cutting attachment has stopped before the unit is set down.

Emissions Control Components

NOTICE

- The use of emission control components other than those specifically designed for this unit is a violation of federal law.
1. Air Filter: Close choke, remove air filter cover, clean air cleaner area, clean or replace filter (if damaged).
 2. Spark Plug: Use only NGK BPM8Y spark plug (BPMR8Y in Canada) otherwise severe engine damage may occur. Adjust spark plug gap by bending outer electrode to a 0.65mm (0.026in) gap.

⚠ DANGER

- Fuel is VERY flammable. Use extreme care when mixing, storing or handling, or serious personal injury may result.
- 1. Fuel Filter replacement: Use a clean rag to remove loose dirt from around fuel cap and empty fuel tank. Pull the fuel filter from the fuel tank. Remove the filter from the line and install the new filter (do not damage fuel line when removing the fuel filter from the tank).

Transportation

⚠ WARNING

Avoid contact with the cutting edges of the blade. Always use extreme care when carrying or handling the equipment. Use the optional blade cover when transported or in storage.

Short-term Storage

⚠ DANGER

- Store unit in a dry, dust free place, out of the reach of children.
- Do not store in enclosure where fuel fumes may accumulate or reach an open flame or spark.

Long-term Storage (Over 30 Days)

NOTICE

- Place the stop switch in the "OFF" position.
- Clean the exterior of the product.
- Perform all periodic maintenance.
- Tighten all the screws and nuts.
- Drain the fuel and run the unit until it stops.
- Allow engine to cool.

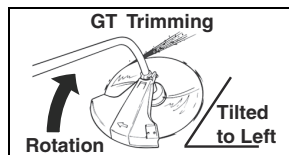
SPECIFIC SAFETY REQUIREMENTS

⚠ WARNING

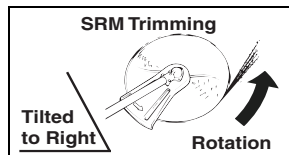
Read and understand all provided literature before use. Failure to do so could result in serious injury.

Cutting Technique

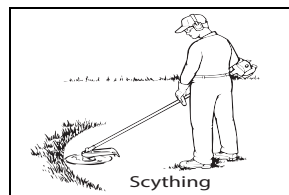
- Nylon line heads may be used for trimming, scything, edging, scalping grass and light weeds.
- GT models: Tilt cutting head to the left while cutting to direct debris away from the operator.



- SRM models: Tilt cutting head to the right while cutting to direct debris away from the operator.



- Either model: To "scythe" swing the cutting head in a level arc feeding the line into the material to be cut.



⚠ WARNING

- The cutting attachment will continue to rotate even after the throttle is released, maintain control of the unit until it has come to a complete stop.
- Avoid contact with blades. Wear gloves to protect hands when handling or maintaining blades. Metal blades are very sharp and can cause severe injuries, even if engine is off and blades are not moving.
- Blade thrust may occur when the spinning blade contacts an object that it does not immediately cut. Following proper cutting techniques will prevent blade thrust.
- Blade thrust can be violent enough to cause the unit and/or operator to be propelled in any direction, and possibly cause operator to lose control of the unit.
- Blade thrust can occur without warning if the blade snags, stalls or binds.
- Blade thrust is more likely to occur in areas where it is difficult to see the material to be cut.
- Blade must be matched with material to be cut.

Recommended product configurations based on material to be cut:

Material to be Cut	Grass (SRM & GT)	Weed/Grass (SRM & GT)	Weed/Grass (SRM)	Brush (<0.5") (SRM)	Clearing (<2.5") (SRM)
Cutting Attachment	Nylon line head Included	Maxi-Cut head, Pro Maxi-Cut head	3 Tooth Blade, 8 Tooth Blade	80 Tooth Blade	22 Tooth Blade
Shield	Shield Included	Included with unit	Included with unit	Included with Blade	Included with Blade
Handle	Handle Included	U-Handle* or Support Handle w/Barrier Bar	U-Handle* or Support Handle w/Barrier Bar	U-Handle* or Support Handle w/Barrier Bar	U-Handle* or Support Handle w/Barrier Bar
Harness	Not Required	Not required	Supplied w/ Kit	Supplied w/ Kit	Supplied w/ Kit

*ANSI standards require brushcutters be equipped with a barrier bar or restrictive harness. U-Handle ensures a higher safety factor.

⚠ WARNING

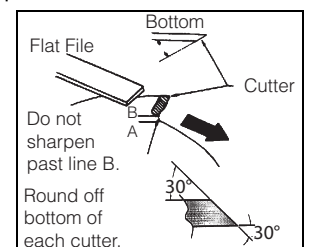
- Blades cannot be used on GT models.
- Use only ECHO approved parts. Failure to use the correct parts can cause the blade to fly off. Seriously injury to the operator and / or bystanders can occur.

Before Each Use When Using A Blade

- Verify the handles are installed according to the manufacturers recommendation.
- Verify the blade is secured properly based on instructions with blade conversion kit.
- Discard blades that are bent, warped, cracked, broken, or damaged in any way.
- Use a sharp blade, dull blades are more likely to snag and thrust.

Sharpening Metal Blades

- Several styles of metal blades are approved for use on the Brushcutter. The 8-tooth blade can be sharpened during normal maintenance. The clearing blade and 80-tooth blade require professional service.
- Before sharpening, CLOSELY inspect blade for cracks (look at the bottom of each tooth and the center mounting hole closely), missing teeth and bending. If ANY of these problems are discovered, replace the blade.
- When sharpening a blade, always remove the same amount of materials from each tooth to maintain balance. A



blade that is not balanced will cause unsafe handling due to vibration and can result in blade failure.

- File each tooth at a 30° angle a specific number of times, e.g., four strokes per tooth. Work your way around the blade until all teeth are sharp.
- DO NOT file the 'gullet' (radius) of the tooth with the flat file. The radius must remain. A sharp corner will lead to a crack and blade failure.
- If an electric grinder is used, use care not to overheat teeth, do not allow tips/tooth to glow red or turn blue. DO NOT place blade in cooling water. This will change the temper of the blade and could result in blade failure.
- After sharpening teeth, check each tooth radius for evidence of a square (sharp) corner. Use the round (rat tail) file to renew the radius.

EMISSION CONTROL (Exhaust & Evaporative)

- CARB And EPA Emissions Control Information.
- The emission control system for the engine is EM (engine modification) and, if the second to last character of the Engine Family on the Emission Control Information label (sample below) is "B", "C", "K", or "T", the emission control system is EM and TWC (3-way catalyst). The fuel tank/fuel line emission control system is EVAP (evaporative emissions). Evaporative emissions for California models are only applicable to fuel tanks and fuel feed lines.

EMISSION CONTROL INFORMATION

ENGINE FAMILY: #EHXS.0364KC DISPLACEMENT: 35.8 cc
EMISSION COMPLIANCE PERIOD: 300 Hours
THIS ENGINE MEETS * * * * U.S. EPA EXH/EVP & CALIFORNIA
EXH/EVP EMISSION REGULATIONS FOR S.O.R.E. REFER TO OWNER'S
MANUAL FOR MAINTENANCE SPECIFICATIONS AND ADJUSTMENTS.
YAMABIKO CORP. * * * X X X X



- An Emission Control Label is located on the engine. (This is an EXAMPLE ONLY, information on label varies by engine FAMILY).
- Product Emission Durability (Emission Compliance Period).
- The 50 or 300 hour emission compliance period is the time span selected by the manufacturer certifying the engine emissions output meets applicable emissions regulations, provided that approved maintenance procedures are followed as listed in the Maintenance Section of this manual.

SERVICE

- Service of this product during the warranty period must be performed by an Authorized ECHO Service Dealer. For the name and address of the Authorized ECHO Service Dealer nearest you, ask your retailer or call 1-800-432-ECHO (3246). Dealer information is also available on our Web Site www.echo-usa.com. When presenting your unit for Warranty service/repairs, proof of purchase is required.

WARNING

Cancer and Reproductive Harm.
www.P65Warnings.ca.gov

Note: This product complies with CAN ICES-2/NMB-2.

MANTENIMIENTO

Pantalla del supresor de chispas

Piezas necesarias: Pantalla del supresor de chispas, Junta

1. Retire el cable de las bujías y la cubierta del motor.
2. Coloque el pistón en punto muerto superior (TDC) para evitar que la suciedad o el carbono entren al cilindro.
3. Retire la cubierta de la pantalla del supresor de chispas, las juntas y la pantalla del cuerpo del silenciador.
4. Limpie los depósitos de carbono de los componentes del silenciador.

Nota: Cuando limpie el depósito de carbono, tenga cuidado de no dañar el elemento catalizador dentro del silenciador (si está equipado con un elemento catalizador).

5. Reemplace la pantalla si está fisurada, obstruida o tiene agujeros quemados de lado a lado.

6. Ensamble los componentes en orden inverso.

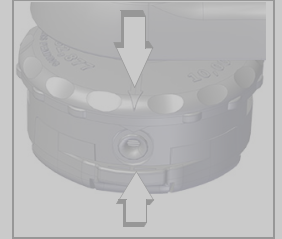
Reemplazo de la línea de nailon

ADVERTENCIA

- Nunca use cable ni cable metálico que pueda desprenderse y convertirse en un "proyector" peligroso. Pueden producirse lesiones graves.
- Use guantes para evitar lesiones personales dado que:
- La cuchilla de corte está afilada.
- La caja de engranajes y sus alrededores pueden estar a alta temperatura.

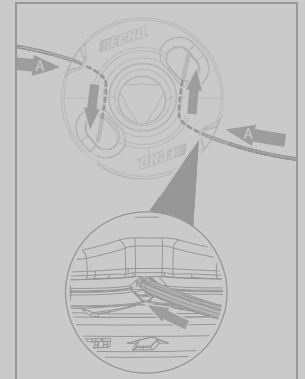
Speed Feed™

1. Corte una pieza de 2 mm (0.80 in) o 2.4 mm (0.95 in) de línea a la longitud recomendada de 6 m (20 ft).
2. Alinee las flechas en la parte superior de la perilla con la abertura de los ojales.
3. Introduzca un extremo de la línea de la podadora en el ojal y presione la línea hasta que longitudes iguales se extiendan del cabezal de la podadora.
4. Sostenga la cabeza del cortacésped y gire la perilla para enrollar la línea en la bobina.
5. Continúe hasta que aproximadamente 13 cm (5 in) de línea se extiendan a cada lado.



Rapid Loader™

1. APAGUE el motor. Coloque la unidad sobre el suelo con el ensamblaje del cabezal hacia arriba.
2. Inserte una pieza de línea de la podadora a través del ojal (A) a cada lado del cabezal. Inserte la línea como se indica en la imagen.
3. Retire la línea de nailon vieja en la dirección que se muestra.



Ajuste de la velocidad de ralentí

AVISO

- Cada unidad se pone en marcha en la fábrica y el carburador se ajusta de acuerdo con las reglamentaciones de emisiones. Los ajustes al carburador, distinto a la velocidad de ralentí, debe realizarlos un distribuidor ECHO autorizado.
- Si hay un tacómetro disponible, el tornillo en velocidad de ralentí (A) (Figura 6A) debe ajustarse según las especificaciones que se encuentran en el Manual del operador. Gire el tornillo de ralentí (A) en sentido horario para aumentar la velocidad de ralentí y en sentido antihorario para disminuirla.

ADVERTENCIA

- Es posible que el accesorio de corte esté girando durante los ajustes del carburador.
- Use el equipo de protección y observe todas las instrucciones de seguridad.
- Para las unidades equipadas con un embrague, asegúrese de que el accesorio de corte deje de girar cuando el motor se mueva en ralentí.
- Cuando la unidad esté apagada, asegúrese de que el accesorio de corte se haya detenido antes de que se asiente la unidad.

Componentes de control de emisiones

AVISO

- El uso de componentes de control de emisiones diferentes de aquellos diseñados específicamente para esta unidad viola la ley federal.
- Filtro de aire: Cierre el estrangulador, retire la cubierta del filtro de aire, limpie el área del limpiador de aire, limpie o reemplace el filtro (si está dañado).
 - Bujía: Utilice solo bujías de encendido NGK BPM8Y (BPMR8Y en Canadá), de lo contrario pueden producirse daños importantes en el motor. Ajuste la separación de la bujía doblando el electrodo exterior para formar una separación de 0.65 mm (0.026 in).

! PELIGRO

- El combustible es MUY inflamable. Tenga extremo cuidado al mezclar, almacenar o manipular el combustible, o pueden producirse graves lesiones personales.
- Reemplazo del filtro de combustible: Use un trapo limpio para eliminar la suciedad suelta alrededor de tapón del combustible y vacíe el tanque de combustible. Retire el filtro de combustible del tanque de combustible. Retire el filtro de la línea e instale el nuevo filtro (no dañe la línea de combustible cuando retire el filtro de combustible del tanque).

Transporte

! ADVERTENCIA

Evite el contacto con los bordes cortantes de la hoja. Tenga extremo cuidado cuando transporte o manipule el equipo. Use la cubierta opcional de la hoja durante el transporte o el almacenamiento.

Almacenamiento a corto plazo

! PELIGRO

- Almacene la unidad en un lugar seco y sin polvo, fuera del alcance de los niños.
- No almacene la unidad en recintos donde los gases del combustible se puedan acumular o alcanzar una llama abierta o una chispa.

Almacenamiento a largo plazo (más de 30 días)

AVISO

- Coloque el interruptor de parada en la posición "OFF" (APAGADO).
- Limpie el exterior del producto.
- Realice todo el mantenimiento periódico.
- Ajuste todos los tornillos y las tuercas.
- Drene el combustible y ponga en marcha la unidad hasta que se detenga.
- Permita que el motor se enfríe.

REQUISITOS DE SEGURIDAD ESPECÍFICOS

! ADVERTENCIA

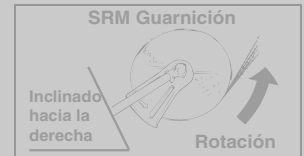
Lea y comprenda toda la bibliografía provista antes del uso. De lo contrario, pueden producirse lesiones graves.

Técnica de corte

- Los cabezales de la línea de nailon pueden usarse para podar, segar, bordear, arrancar pastos y pequeñas hierbas.
- Modelos GT: Inclina la cabeza de corte hacia la izquierda mientras corta para dirigir la suciedad lejos del operador.



- Modelos SRM: Inclina la cabeza de corte hacia la derecha mientras corta para dirigir los desechos lejos del operador.



- Cualquier modelo: Para "segar" balancee el cabezal de corte en un arco plano que alimente la línea al material que debe cortarse.



! ADVERTENCIA

- El accesorio de corte continuará girando aun después de que se libere el acelerador. Mantenga el control de la unidad hasta que se haya detenido por completo.
- Evite el contacto con las hojas. Use guantes para proteger las manos cuando manipule las hojas o realice su mantenimiento. Las hojas de metal son muy filosas y pueden causar lesiones graves, incluso si el motor está apagado y las hojas no se están moviendo.
- Pueden producirse contragolpes cuando la hoja que está girando entra en contacto con un objeto que no corta inmediatamente. Si se siguen las técnicas de corte adecuadas, se evitarán los contragolpes.
- Los contragolpes pueden ser suficientemente violentos para provocar que la unidad o el operador sean impulsados en cualquier dirección y, posiblemente, que el operador pierda el control de la unidad.
- Los contragolpes pueden producirse sin advertencia si la hoja se engancha, se detiene o se enlaza.
- Es más probable que se produzcan contragolpes en áreas donde es difícil ver el material que se corta.
- La hoja debe ser apta para el material que se corta.

Configuraciones de producto recomendadas según el material que se cortará:

Material para cortar	Césped (SRM y GT)	Hierbas/ Césped (SRM y GT)	Hierbas/ Césped (SRM)	Desmalezad ora (<0.5") (SRM)	Limpieza (<2.5") (SRM)
Accesorio de corte	Cabezal de línea de nailon incluido	Cabezal de maxicorte, Cabezal de maxicorte pro	Hoja de 3 dientes, Hoja de 8 dientes	Hoja de 80 dientes	Hoja de 22 dientes
Placa	Placa incluida	Incluido con la unidad	Incluido con la unidad	Incluido con la hoja	Incluido con la hoja
Mango	Mango incluido	Mango en U* o mango de soporte Con barra de barrera	Mango en U* o mango de soporte Con barra de barrera	Mango en U* o mango de soporte Con barra de barrera	Mango en U* o mango de soporte Con barra de barrera
Arnés	No es necesario	No es necesario	Se proporciona con el kit	Se proporciona con el kit	Se proporciona con el kit

*Los estándares ANSI requieren que las desmalezadoras estén equipadas con una barra de barrera o un arnés restrictivo. El mango en U garantiza un mayor factor de seguridad.

! ADVERTENCIA

- Las hojas no pueden usarse en los modelos GT.
- Use solo piezas ECHO aprobadas. Si no usa las piezas correctas, la hoja puede salir volando. Pueden producirse lesiones graves para el operador o las demás personas presentes.

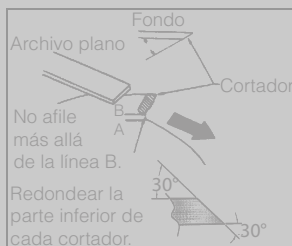
Antes de cada uso cuando se usa una hoja

- Verifique que los mangos estén instalados según las recomendaciones de los fabricantes.
- Verifique que la hoja esté asegurada correctamente según las instrucciones con el kit de conversión de hoja.

- Descarte las hojas que estén dobladas, arqueadas, rajadas, rotas o dañadas de alguna forma.
- Use una hoja filosa, ya que las hojas romas son más propensas a engancharse y causar contragolpes.

Afilar hojas metálicas

- Hay varios estilos de hojas metálicas aprobados para usar con la desmalezadora. La hoja de 8 dientes puede afilarse durante el mantenimiento normal. La hoja de limpieza y la hoja de 80 dientes requieren mantenimiento profesional.
- Antes del afilado, inspeccione la hoja DETALLADAMENTE para ver si hay rajaduras (observe el inferior de cada diente y el orificio del montaje central detenidamente), dientes faltantes y curvaturas. Si descubre CUALQUIERA de estos problemas, reemplace la hoja.
- Cuando afile una hoja, retire siempre la misma cantidad de materiales de cada diente para mantener el equilibrio. Una hoja que no está equilibrada da lugar a una manipulación insegura debido a la vibración y puede dar como resultado una falla de la hoja.
- Lime cada diente a un ángulo de 30° una cantidad específica de veces, por ej., cuatro veces por diente. Avance alrededor de la hoja hasta que todos los dientes queden filosos.
- NO lime la 'garganta' (el radio) del diente con la lima plana. El radio debe permanecer igual. Una esquina filosa ocasionará una rajadura y una falla de la hoja.
- Si se usa un afilador eléctrico, tenga cuidado de no sobrecalentar los dientes, y no permita que las puntas ni el diente brillen en rojo ni se tornen azules. NO coloque la hoja en agua helada. Esto cambiará la temperatura de la hoja y puede provocar una falla de la hoja.
- Después de afilar los dientes, verifique cada radio de los dientes para detectar evidencia de una esquina cuadrada (filosa). Use la lima redonda (con cola de rata) para renovar el radio.



CONTROL DE EMISIONES (escape y evaporativas)

- Información sobre el control de emisiones de CARB y EPA.
- El sistema de control de emisiones del motor es EM (modificación del motor) y, si el penúltimo carácter de la familia de motores que aparece en la etiqueta de información sobre el control de emisiones (ejemplo a continuación) es "B", "C", "K" o "T", el sistema de control de emisiones es EM y TWC (catalizador de 3 vías). El sistema de control de emisiones del tanque y tubería de combustible es EVAP (emisiones evaporativas). Las emisiones evaporativas para los modelos de California solo se aplican a tanques de combustible y a líneas de alimentación de combustible.

EMISSION CONTROL INFORMATION

ENGINE FAMILY: #EHXS.0364KC DISPLACEMENT: 35.8 cc
EMISSION COMPLIANCE PERIOD: 300 Hours
THIS ENGINE MEETS * * * * U.S. EPA EXH/EVP & CALIFORNIA
EXH/EVP EMISSION REGULATIONS FOR S.O.R.E. REFER TO OWNER'S
MANUAL FOR MAINTENANCE SPECIFICATIONS AND ADJUSTMENTS.
YAMABIKO CORP. * * * X X X X

- Puede encontrarse una etiqueta de control de emisiones en el motor. (Este es SOLO UN EJEMPLO, la información en la etiqueta varía según la FAMILIA de motor).
- Durabilidad de las emisiones del producto (período de cumplimiento de las emisiones)
- El período de cumplimiento de las emisiones de 50 o 300 horas es el intervalo de tiempo seleccionado por el fabricante por el que se certifica que la salida de emisiones del motor cumple con las reglamentaciones correspondientes, siempre que se sigan los procedimientos de mantenimiento aprobados según lo establecido en la sección Mantenimiento de este manual.

MANTENIMIENTO

- Durante el período de garantía, el mantenimiento de este producto lo debe realizar un distribuidor de mantenimiento de ECHO autorizado. Para conocer el nombre y el domicilio del distribuidor de mantenimiento de ECHO autorizado más cercano, pregúnteselo a su revendedor o llame al 1-800-432-ECHO (3246). La información sobre el distribuidor también está disponible en nuestro sitio web www.echo-usa.com. Cuando presente su unidad para que se le realicen mantenimiento o reparaciones bajo la garantía, se requiere mostrar prueba de compra.

! ADVERTENCIA

Cáncer y daño reproductivo
www.P65Warnings.ca.gov

Nota: Este producto cumple con los estándares CAN ICES-2/NMB-2.

ENTRETIEN

Écran pare-étincelles

Pièces requises : Écran pare-étincelles, garniture

1. Retirer le câble de la bougie et le capot du moteur.
2. Placer le piston au point mort haut (PMH) afin d'empêcher l'entrée du carbone/la saleté dans le cylindre.
3. Retirer le couvre-écran du pare-étincelles, les garnitures et l'écran du silencieux.
4. Essuyer les résidus de carbone des composants du silencieux.

Remarque: En nettoyant les résidus de carbone, faire attention; ne pas endommager l'élément catalyseur du silencieux (si équipé d'un élément catalyseur).

5. Remplacer l'écran s'il est brisé, bouché ou a des trous qui ont brûlés en travers.
6. Assembler les composants dans l'ordre inverse.

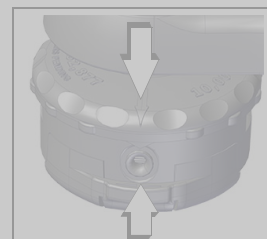
Remplacement du câble en nylon

! AVERTISSEMENT

- Ne jamais utiliser du fil ou un câble qui peut se couper et devenir un « projectile » dangereux. Des blessures graves peuvent survenir.
- Porter des gants, sinon des blessures graves peuvent survenir :
- La lame de coupe est tranchante.
- Le boîtier d'engrenage et les environs peuvent surchauffer.

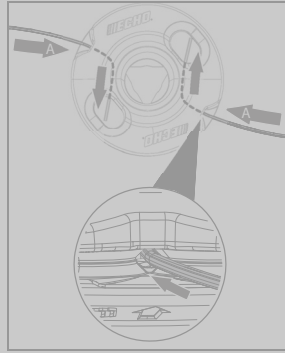
Speed Feed™

1. Couper un bout de 2,0 mm (0,80 po) ou 2,4 mm (0,95 po) de fil pour la longueur recommandée de 6 m (20 pi).
2. Aligner les flèches sur la manette avec ouverture d'œillets.
3. Insérer une extrémité du coupe-herbe dans l'œillet et pousser le fil jusqu'à ce que la longueur soit égale de part et d'autre du coupe-herbe.
4. Tenez la tête de la tondeuse et tournez le bouton pour enrouler le fil sur la canette.
5. Continuer jusqu'à ce qu'il y a environ 13 cm (5 po) de fil de chaque côté.



Rapid Loader™

1. Arrêter le moteur. Poser le dispositif sur le sol avec la tête vers le haut.
2. Insérer un bout de fil du coupe-herbe dans l'œillet (A) de part et d'autre de la tête. L'itinéraire à suivre est tel qu'indiqué.
3. Retirer le fil en nylon usé dans la direction indiquée.



Réglage de vitesse de ralenti

AVIS

- Chaque dispositif est testé à l'usine et le carburateur est réglé conformément aux règlements sur les émissions. Les réglages de carburateur, à part la vitesse de ralenti, doivent être effectués par un détaillant agréé ECHO.
- Si un tachymètre est disponible, la vis (A) de vitesse du ralenti (Figure 6A) doit être ajustée selon les spécifications prescrites dans le manuel de l'utilisateur. Tourner la vis de vitesse de ralenti (A) dans le sens horaire pour augmenter la vitesse de ralenti, et dans le sens antihoraire pour diminuer la vitesse de ralenti.

AVERTISSEMENT

- L'accessoire de coupe peut tourner pendant les réglages sur le carburateur.
- Porter l'équipement de sécurité et suivre toutes les instructions de sûreté.
- Pour les dispositifs équipés d'un embrayage, s'assurer que l'accessoire de coupe s'arrête de tourner lorsque le moteur est au ralenti.
- Lorsque le dispositif est éteint, s'assurer que l'accessoire de coupe s'est arrêté avant poser le dispositif sur le sol.

Éléments antipollution

AVIS

- L'utilisation d'éléments antipollution différents de ceux qui ont été spécialement conçus pour ce dispositif constitue une infraction à la loi fédérale.
1. Filtre à air : Volet fermé; retirer le capot du filtre à air; air propre, zone plus propre; nettoyer ou remplacer le filtre (si endommagé).
 2. Bougie d'allumage : N'utiliser que la bougie NGK BPM8Y (BPMR8Y au Canada); le cas contraire pourrait sérieusement endommager le moteur. Régler l'écart de bougie d'allumage en pliant l'électrode extérieur à un écart de 0,65 mm (0,026 po).

DANGER

- L'essence est TRÈS inflammable. Faire très attention pendant le mélange, la conservation et la manipulation; sinon des blessures graves peuvent survenir.

1. Remplacement du filtre à carburant : Utiliser un chiffon propre pour essuyer la saleté autour du bouchon du réservoir et le réservoir vide. Ôter le filtre à carburant du réservoir à carburant. Retirer le filtre du fil et installer un nouveau (ne pas endommager la conduite de carburant en retirant filtre à carburant du réservoir).

Transport

AVERTISSEMENT

Éviter tout contact avec les bouts tranchants de la lame. Faire extrêmement attention lors du transport ou de la manipulation de l'équipement. Utiliser le couvre-lame facultatif pendant le transport ou le rangement du dispositif.

Rangement à court terme

DANGER

- Ranger le dispositif dans un endroit sec, à l'abri de la poussière et hors de portée des enfants.
- Ne pas ranger dans une enceinte où les émissions de carburant peuvent s'accumuler et atteindre une flamme ou une étincelle.

Rangement à long terme (Plus de 30 jours)

AVIS

- Placer l'interrupteur d'arrêt en position « OFF ».
- Nettoyer l'extérieur du produit.
- Effectuer toutes les procédures de maintenance périodique.
- Resserer toutes les vis et écrous.
- Drainer le carburant et démarrer le dispositif jusqu'à ce qu'il s'arrête de marcher.
- Laisser le moteur se refroidir.

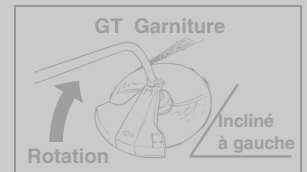
EXIGENCES DE SÉCURITÉ PARTICULIÈRES

AVERTISSEMENT

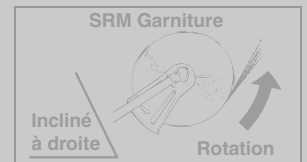
Lire et comprendre toute la documentation à votre disposition avant utilisation. Le non-respect de ces instructions peut entraîner de graves blessures.

Technique de coupe

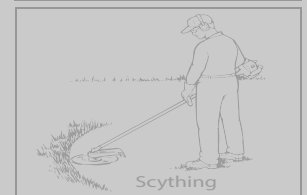
- Les têtes de nylon peuvent servir à la coupe, au fauchage, au raclage et au scalpage d'herbes et d'algues légères.
- Modèles GT : Pencher la tête coupante vers la gauche pour diriger les débris à l'écart de l'utilisateur.



- Modèles SRM : Pencher la tête coupante vers la droite, pour diriger les débris à l'écart de l'utilisateur.



- L'un ou l'autre modèle : Pour « faucher » balancer la tête coupante en formant un arc droit, en orientant la ligne de coupe vers le matériau à couper.



AVERTISSEMENT

- L'accessoire de coupe va continuer de tourner même après relâchement du câble d'accélération. Continuer de contrôler le dispositif jusqu'à l'arrêt total du moteur.
- Éviter le contact avec les lames. Porter les gants pour protéger les mains pendant la manipulation et le maintien des lames. Les lames métalliques sont très tranchantes et peuvent causer des blessures graves, même lorsque le moteur est éteint et les lames ne bougent plus.
- Le ricochet de lame peut survenir lorsque la lame tournante entre en contact avec un objet qu'elle ne coupe pas aussitôt. Suivre les techniques adéquates de coupe va prévenir le ricochet de lame.

- La violence d'un ricochet de lame peut entraîner la propulsion de l'appareil et/ou de l'opérateur dans n'importe quelle direction, ce qui peut entraîner la perte de contrôle de l'appareil.
- Le ricochet de lame peut se produire sans avertissement si la lame s'accroche, se décroche ou se lie.
- Le ricochet de lame arrive très souvent dans les zones où l'on a du mal à voir le matériau à couper.
- La lame doit correspondre au matériau à couper.

Les configurations recommandées de produits sont fonction du matériau à couper :

Matériau à couper	Herbe (SRM et GT)	Algue/herbe (SRM et GT)	Algue/herbe (SRM)	Débroussaillieuse (<0,5 po) (SRM)	Défrichage (<2,5 po) (SRM)
Accessoire de coupe	Tête de conduite en nylon Inclusive	Tête Maxi-Coupe, Tête Pro Maxi-Coupe	Lame à 3 dents, Lame à 8 dents	Lame à 80 dents	Lame à 22 dents
Capot	Capot inclus	Inclus avec le dispositif	Inclus avec le dispositif	Inclus avec la lame	Inclus avec la lame
Poignée	Avec poignée	Poignée en U * ou poignée de soutien Avec barre de protection	Poignée en U * ou poignée de soutien Avec barre de protection	Poignée en U * ou poignée de soutien Avec barre de protection	Poignée en U * ou poignée de soutien Avec barre de protection
Harnais	Non Exigé	Non exigé	Fourni avec la trousse	Fourni avec la trousse	Fourni avec la trousse

* Les normes ANSI exigent que les débroussaillieuses soient équipées d'une barre de protection ou d'un harnais restrictif. La poignée en U est un facteur de sécurité plus sûr.



AVERTISSEMENT

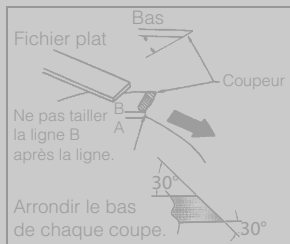
- Les lames ne peuvent pas être utilisées sur les modèles GT.
- N'utiliser que des pièces approuvées par ECHO. Si vous n'utilisez pas les bonnes pièces, la lame risque de s'envoler. Des blessures graves pour l'opérateur et/ou les personnes se trouvant à proximité peuvent survenir.

Avant chaque utilisation d'une lame

- Vérifier que les poignées sont installées conformément aux recommandations du fabricant.
- Vérifier que la lame est fixée correctement en suivant les instructions de la trousse de conversion de lame.
- Jeter les lames pliées, déformées, fissurées, voilées, cassées ou endommagées de quelque façon que ce soit.
- Utiliser une lame tranchante, les lames émoussées sont plus susceptibles de s'accrocher et de ricocher.

Affûtage des lames métalliques

- Plusieurs modèles de lames métalliques sont approuvés pour utilisation sur la débroussaillieuse. La lame à 8 dents peut être affûtée pendant le processus d'entretien normal. La lame de défrichage et la lame à 80 dents doivent faire l'objet d'un entretien professionnel.
- Avant l'affûtage, inspecter MÉTICULEUSEMENT la lame en cas de fissures (regarder de près le fond de chaque dent et le trou de montage central), de dents manquantes et de courbure. Si l'un de ces défauts est détecté, remplacer la lame.
- Pendant l'affûtage d'une lame, toujours ôter la même quantité de matériau de chaque dent pour maintenir l'équilibre. Une lame qui n'est pas équilibrée causera une manipulation dangereuse en raison des vibrations et peut entraîner une défaillance de la lame.
- Limer chaque dent à un angle de 30° un certain nombre de fois, par exemple, quatre coups par dent. Faites le tour de la lame jusqu'à ce que toutes les dents soient aiguisées.
- NE PAS limer la rainure (rayon) de la dent avec la lime plate. Le rayon doit rester intact. Un coin aigu peut entraîner une fissure et une défaillance de la lame.



CONTRÔLE DES ÉMISSIONS (Échappement et évaporation)

- Informations sur le contrôle des émissions du CARB et de l'EPA.
- Le système antipollution du moteur est EM (modification du moteur) et, si l'avant-dernier caractère de la famille de moteurs sur l'étiquette d'informations antipollution (échantillon ci-dessous) est « B », « C », « K » ou « T », le système antipollution est EM et TWC (catalyseur à 3 voies). Le système de contrôle des émissions du réservoir de carburant/ de la conduite de carburant est EVAP (émissions par évaporation). Les émissions par évaporation des modèles californiens ne s'appliquent qu'aux réservoirs de carburant et aux conduites d'alimentation en carburant.

EMISSION CONTROL INFORMATION

ENGINE FAMILY: #EHXS.0364KC DISPLACEMENT: 35.8 cc
EMISSION COMPLIANCE PERIOD: 300 Hours
THIS ENGINE MEETS * * * * U.S. EPA EXH/EVP & CALIFORNIA
EXH/EVP EMISSION REGULATIONS FOR S.O.R.E. REFER TO OWNER'S
MANUAL FOR MAINTENANCE SPECIFICATIONS AND ADJUSTMENTS.
YAMABIKO CORP. * * * X X X X



- Une étiquette d'information sur la réduction des émissions est apposée sur le moteur. (Ceci N'EST QU'UN SPÉCIMEN, l'information sur l'étiquette varie selon la FAMILLE de moteur).
- Durabilité des émissions du produit (période de conformité des émissions).
- La période de conformité de 50 ou 300 heures est la période de temps choisie par le fabricant pour certifier que les émissions du moteur sont conformes aux règlements sur les émissions applicables, à condition que les procédures d'entretien approuvées soient suivies comme indiqué dans la section Entretien du présent manuel.

SERVICE

- L'entretien de ce produit pendant la période de garantie doit être effectué par un distributeur agréé ECHO. Pour obtenir le nom et l'adresse du distributeur agréé ECHO le plus près de chez vous, demandez à votre revendeur ou appelez le 1 800 432-ECHO (3246). Des renseignements sur les concessionnaires sont également disponibles sur notre site Web www.echo-usa.com. Lorsque vous présentez votre appareil pour un service ou une réparation sous garantie, une preuve d'achat est requise.

**AVERTISSEMENT**

Cancer et troubles de la reproduction.
www.P65Warnings.ca.gov

Remarque: Ce produit est conforme à CAN ICES-2 / NMB-2.